

Dimensionality Reduction the Probabilistic Way

An Overview of Probabilistic Dimensionality Reduction

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ICML Tutorial
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- 1 Thinking in High Dimensions
- 2 Probabilistic Dimensionality Reduction
- 3 Spectral Methods and Probabilistic Approaches
- 4 Density Networks and GTM
- 5 Dual Probabilistic PCA and GP-LVM
- 6 Applications
- 7 Conclusions

All source code and slides are available online

- Tutorial homepage is
 - ▶ http://www.cs.man.ac.uk/~neill/icml_tutorial.html.
 - ▶ MATLAB/Octave commands used for examples given in typewriter font.

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Mixtures of Gaussians

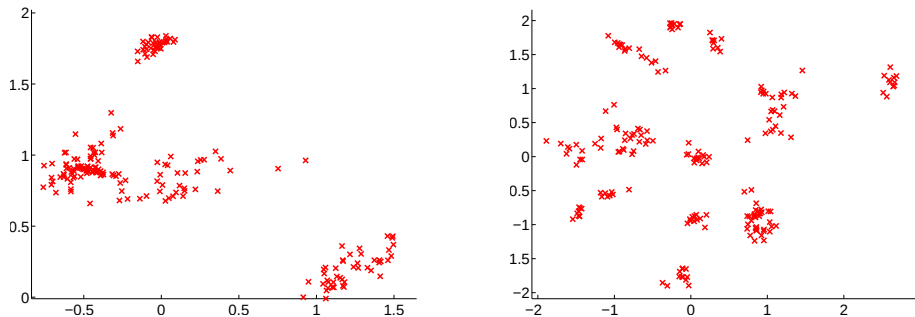


Figure: Two dimensional data sets.

Mixtures of Gaussians

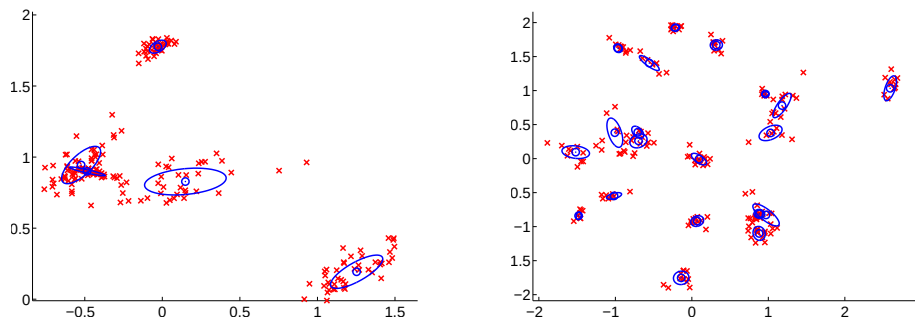


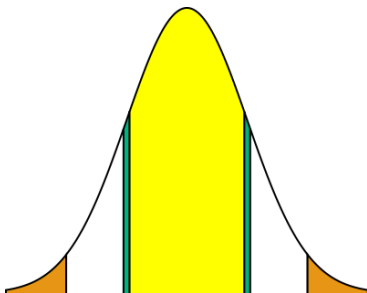
Figure: Complex structure not a problem for mixtures of Gaussians.

Thinking in High Dimensions

- Two dimensional plots of Gaussians can be misleading.
- Our low dimensional intuitions can fail dramatically.
- Two major issues:
 - 1 In high dimensions all the data moves to a 'shell'. There is nothing near the mean!
 - 2 Distances between points become constant (this applies for many distributions).
- Let's consider a Gaussian "egg".

The Gaussian Egg

See also Exercise 1.4 in (Bishop, 1995)

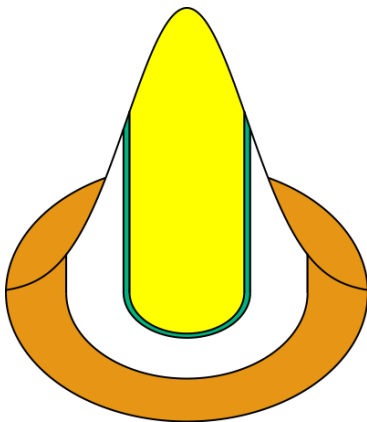


Volumes: 65.8%, 4.84%, 24.82%, 4.24%

Figure: One dimensional Gaussian distribution.

The Gaussian Egg

See also Exercise 1.4 in (Bishop, 1995)

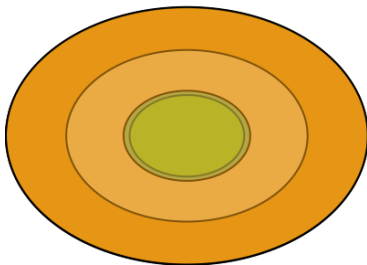


Volumes: 59.4%, 7.35% 31.37% 1.82%

Figure: Two dimensional Gaussian distribution.

The Gaussian Egg

See also Exercise 1.4 in (Bishop, 1995)



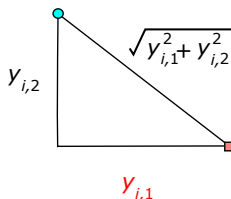
Volumes: 56.1%, 9.24%, 33.92% 0.74%

Figure: Three dimensional Gaussian distribution.

What is the distribution of mass?

$$y_{i,k} \sim \mathcal{N}(0, \sigma^2)$$

$$\implies y_{i,k}^2 \sim \sigma^2 \chi_1^2$$

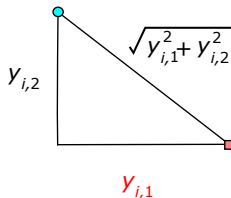


Square of sample from Gaussian is scaled chi-squared distribution

What is the distribution of mass?

$$y_{i,k} \sim \mathcal{N}(0, \sigma^2)$$

$$\Rightarrow y_{i,k}^2 \sim \mathcal{G}\left(\frac{1}{2}, \frac{1}{2\sigma^2}\right)$$

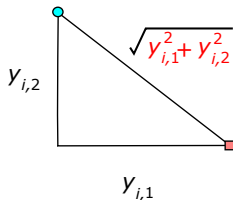


Chi squared distribution is a variant of the gamma distribution with shape parameter $a = \frac{1}{2}$, rate parameter $b = \frac{1}{2\sigma^2}$, $\mathcal{G}(x|a, b) = \frac{b^a}{\Gamma(a)} x^{a-1} e^{-bx}$.

What is the distribution of mass?

$$y_{i,k} \sim \mathcal{N}(0, \sigma^2)$$

$$\Rightarrow y_{i,1}^2 + y_{i,2}^2 \sim \mathcal{G}\left(\frac{2}{2}, \frac{1}{2\sigma^2}\right)$$

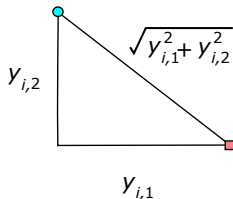


Addition of gamma random variables with the same rate is gamma with sum of shape parameters ($y_{i,k}$ s are independent)

What is the distribution of mass?

$$\sum_{k=1}^D y_{i,k}^2 \sim \mathcal{G}\left(\frac{D}{2}, \frac{1}{2\sigma^2}\right)$$

$$\Rightarrow \left\langle \sum_{k=1}^D y_{i,k}^2 \right\rangle = D\sigma^2$$

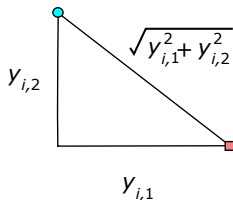


Addition of gamma random variables with the same rate is gamma with sum of shape parameters ($y_{i,k}$ s are independent)

What is the distribution of mass?

$$\frac{1}{D} \sum_{k=1}^D y_{i,k}^2 \sim \mathcal{G} \left(\frac{D}{2}, \frac{D}{2\sigma^2} \right)$$

$$\Rightarrow \left\langle \frac{1}{D} \sum_{k=1}^D y_{i,k}^2 \right\rangle = \sigma^2$$



Scaling of gamma distribution scales the rate parameter

Where is the Mass?

- Squared distances from mean are gamma distributed.
- Can use gamma density's cumulative distribution to work out where the mass is.

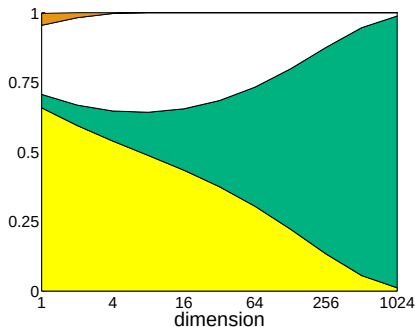


Figure: Plot of probability mass versus dimension. Plot shows the volume of distribution inside 0.95 of a standard deviation (yellow), between 0.95 and 1.05 standard deviations (green), between 1.05 and 2 standard deviations (white) and between 2 and 3 standard deviations (brown).

Looking at Gaussian Samples

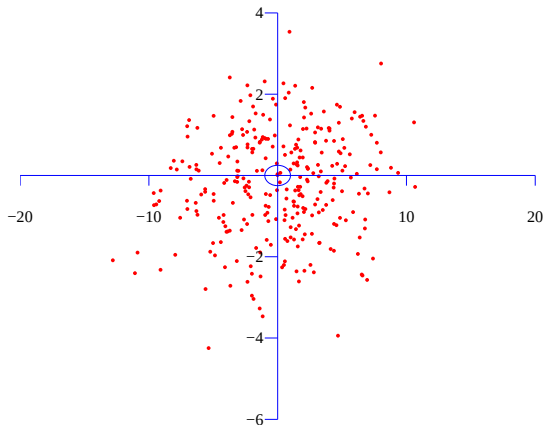


Figure: How to look at a projected Gaussian.

Interpoint Distances

- The other effect in high dimensions is all points become equidistant.
- Can show this for Gaussians with a similar proof to the above,

$$y_{i,k} \sim \mathcal{N}(0, \sigma_k^2) \quad y_{j,k} \sim \mathcal{N}(0, \sigma_k^2)$$

$$y_{i,k} - y_{j,k} \sim \mathcal{N}(0, 2\sigma_k^2)$$

$$(y_{i,k} - y_{j,k})^2 \sim \mathcal{G}\left(\frac{1}{2}, \frac{1}{4\sigma_k^2}\right)$$

For spherical Gaussian, $\sigma_k^2 = \sigma^2$

$$\sum_{k=1}^D (y_{i,k} - y_{j,k})^2 \sim \mathcal{G}\left(\frac{D}{2}, \frac{1}{4\sigma^2}\right)$$

$$\frac{1}{D} \sum_{k=1}^D (y_{i,k} - y_{j,k})^2 \sim \mathcal{G}\left(\frac{D}{2}, \frac{D}{4\sigma^2}\right)$$

Dimension normalized distance between points is Gamma distributed.
Mean is $2\sigma^2$. Variance is $\frac{8\sigma^2}{D}$.

Central Limit Theorem and Non-Gaussian Case

- We could compute these distribution of squared distance *analytically* for spherical, independent Gaussian data.
- More generally, for *independent* data, the **central limit theorem** applies.
 - ▶ The mean squared distance in high dimensional space is the mean of the variances.
 - ▶ The variance about the mean scales as D^{-1} .

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- In high dimensions if individual dimensions are *independent* the distributions behave counter intuitively.
- All data sits at one standard deviation from the mean.
- The distribution of data can be analytically calculated for the Gaussian case.
- For non-Gaussian *independent* systems we can invoke the central limit theorem.
- Next we will consider example data sets and see how their interpoint distances are distributed.

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Sanity Check: Data sampled from independent Gaussian distribution

- If dimensions are independent, we expect low variance, Gaussian behaviour for the distribution of squared distances.

Distance distribution for a Gaussian with $D = 1000$, $N = 1000$

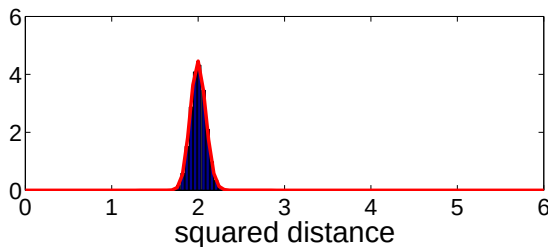


Figure: A good match between theory and the samples for a 1000 dimensional Gaussian distribution.

Same data generation, but fewer data points.

- If dimensions are independent, we expect low variance, Gaussian behaviour for the distribution of squared distances.

Distance distribution for a Gaussian with $D = 1000$, $N = 100$

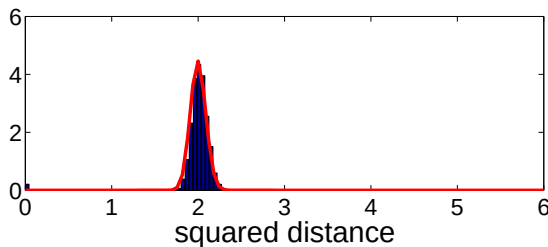
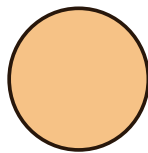


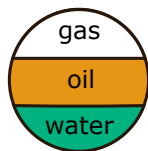
Figure: A good match between theory and the samples for a 1000 dimensional Gaussian distribution.

- Simulated measurements from an oil pipeline (Bishop and James, 1993).
- Pipeline contains oil, water and gas.
- Three phases of flow in pipeline — homogeneous, stratified and annular.
- Gamma densitometry sensors arranged in a configuration around pipeline.

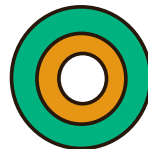
Homogeneous



Stratified

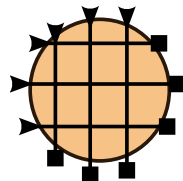


Annular

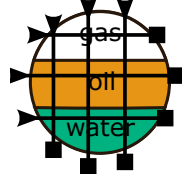


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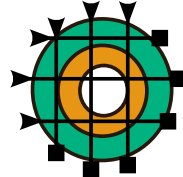
Homogeneous



Stratified



Annular



- 12 simulated measurements of oil flow in a pipe.
- Nature of flow is dependent on relative proportion of oil, water and gas.

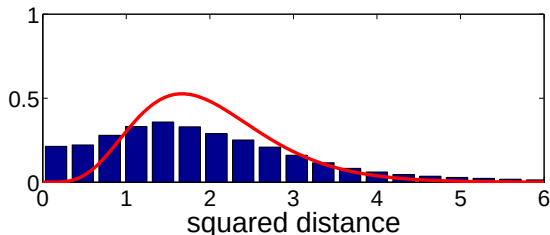


Figure: Interpoint squared distance distribution for oil data with $D = 12$.
(variance of squared distances is 1.98 vs predicted 0.67)

Stick Man Data

- $N = 55$ frames of motion capture.
- xyz locations of 34 points on the body.
- $D = 102$ dimensional data.
- “Run 1” available from http://accad.osu.edu/research/mocap/mocap_data.htm.

Changing



Angle



of Run



- Motion capture data inter point distance histogram.

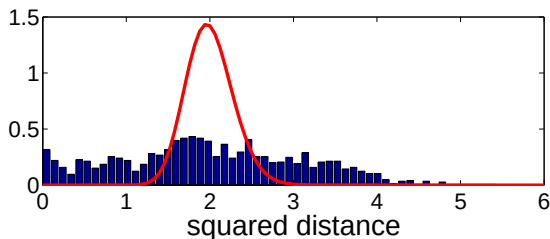


Figure: Interpoint squared distance distribution for stick man data with $D = 102$ (variance of squared distances is 1.09 vs predicted 0.078).

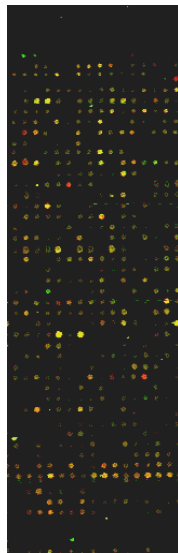
Microarray Data

- Gene expression measurements reflecting the cell cycle in yeast (Spellman et al., 1998).
- $D = 24,401$ Genes measured at $N = 24$ time points
- Data available from <http://genome-www.stanford.edu/cellcycle/data/rawdata/individual.htm>.

Yeast

Cell

Cycle



- Spellman yeast cell cycle.

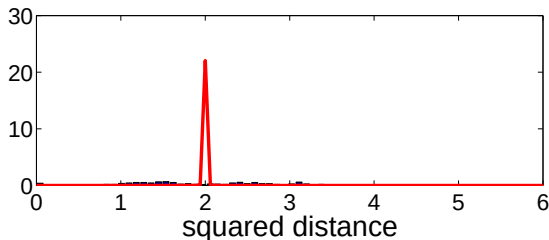


Figure: Interpoint squared distance distribution for Microarray data with $D = 24,401$ (variance of squared distances is 0.6173 vs predicted $3.3\text{e-}4$).

Where does practice depart from our theory?

- The situation for real data does not reflect what we expect.
- Real data exhibits greater variances on interpoint distances.
 - ▶ Somehow the real data seems to have a smaller effective dimension.
- Let's look at another $D = 1000$.

Distance distribution for a different Gaussian with $D = 1000$

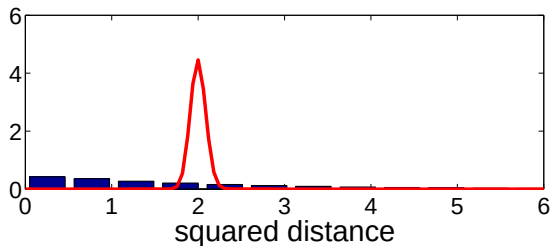


Figure: Interpoint squared distance distribution for Gaussian with $D = 1000$.

Distance distribution for a different Gaussian with $D = 1000$

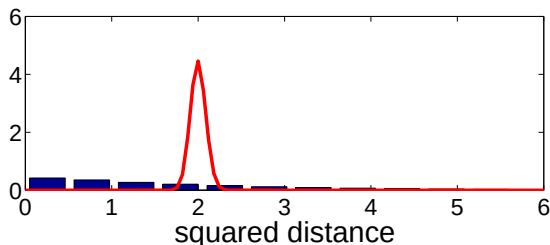


Figure: Interpoint squared distance distribution for Gaussian with $D = 1000$.

Gaussian has a specific low rank covariance matrix $\mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}$.
Take $\sigma^2 = 1e - 2$ and sample $\mathbf{W} \in \mathbb{R}^{1000 \times 2}$ elements independently from $\mathcal{N}(0, 1)$.

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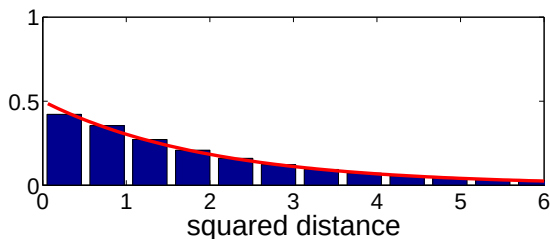


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Where does this Low Rank Covariance Matrix Come From?

- Factor Analysis leads to $\mathbf{C} = \mathbf{W}\mathbf{W}^T + \mathbf{D}$, $\mathbf{D} = \text{diag}(\mathbf{d})$ — not discussed further.
- Probabilistic PCA (Tipping and Bishop, 1999b; Roweis, 1998)
 - ▶ Linear Mapping from q -dimensional latent space to D -dimensional data space.
 - ▶ Corrupt the mapping by independent Gaussian noise.
 - ▶ Marginalise latent variables using Gaussian prior.

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q — dimension of latent/embedded space

D — dimension of data space

N — number of data points

centred data, $\mathbf{Y} = [\mathbf{y}_{1,:}, \dots, \mathbf{y}_{N,:}]^T = [\mathbf{y}_{:,1}, \dots, \mathbf{y}_{:,D}] \in \mathbb{R}^{N \times D}$

latent variables, $\mathbf{X} = [\mathbf{x}_{1,:}, \dots, \mathbf{x}_{N,:}]^T = [\mathbf{x}_{:,1}, \dots, \mathbf{x}_{:,q}] \in \mathbb{R}^{N \times q}$

mapping matrix, $\mathbf{W} \in \mathbb{R}^{D \times q}$

$\mathbf{a}_{i,:}$ is a vector from the i th row of a given matrix $\mathbf{A}$

$\mathbf{a}_{:,j}$ is a vector from the j th row of a given matrix $\mathbf{A}$

$\mathbf{X}$ and $\mathbf{Y}$ are *design matrices*

- Data covariance given by $N^{-1}\mathbf{Y}^T\mathbf{Y}$

$$\text{cov}(\mathbf{Y}) = \frac{1}{N} \sum_{i=1}^N \mathbf{y}_{i,:} \mathbf{y}_{i,:}^T = \frac{1}{N} \mathbf{Y}^T \mathbf{Y}$$

- Inner product matrix given by $\mathbf{Y}\mathbf{Y}^T$

$$\mathbf{K} = (k_{i,j})_{i,j}, \quad k_{i,j} = \mathbf{y}_{i,:}^T \mathbf{y}_{j,:}$$

Linear Dimensionality Reduction

- Two dimensional plane projected into a three dimensional space.

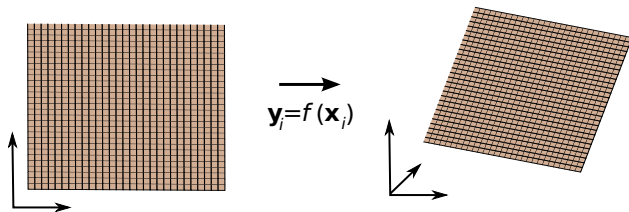


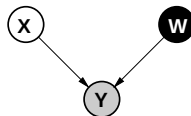
Figure: Mapping a two dimensional plane to a higher dimensional space in a linear way.

Linear Latent Variable Model

- Represent data, $\mathbf{Y}$, with a lower dimensional set of latent variables $\mathbf{X}$.
- Assume a linear relationship of the form

Probabilistic PCA

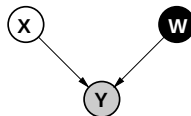
- Linear-Gaussian relationship between latent variables and data.
- $\mathbf{X}$ are 'nuisance' variables.
- Latent variable model approach:
 - ▶ Define Gaussian prior over *latent space*, $\mathbf{X}$.
 - ▶ Integrate out nuisance *latent variables*.



$$p(\mathbf{Y}|\mathbf{X}, \mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{W}\mathbf{x}_{i,:}, \sigma^2 \mathbf{I})$$

Probabilistic PCA

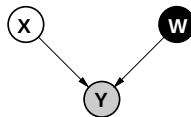
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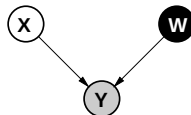
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Probabilistic PCA

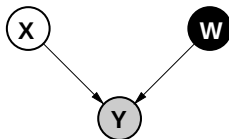
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Probabilistic PCA Max. Likelihood Soln (Tipping and Bishop, 1999b)



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Probabilistic PCA Max. Likelihood Soln (Tipping and Bishop, 1999b)

$$p(\mathbf{Y}|\mathbf{W}) = \prod_{j=1}^D \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{0}, \mathbf{C}), \quad \mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}$$

$$\log p(\mathbf{Y}|\mathbf{W}) = -\frac{N}{2} \log |\mathbf{C}| - \frac{1}{2} \text{tr}(\mathbf{C}^{-1} \mathbf{Y}^T \mathbf{Y}) + \text{const.}$$

If $\mathbf{U}_q$ are first q principal eigenvectors of $N^{-1} \mathbf{Y}^T \mathbf{Y}$ and the corresponding eigenvalues are Λ_q ,

$$\mathbf{W} = \mathbf{U}_q \mathbf{L} \mathbf{R}^T, \quad \mathbf{L} = (\Lambda_q - \sigma^2 \mathbf{I})^{\frac{1}{2}}$$

where $\mathbf{R}$ is an arbitrary rotation matrix.

► Details

Why Probabilistic PCA?

- What is the point in probabilistic methods?
- Could we not just project with regular PCA?
 - ▶ Integration within other models (e.g. mixtures of PCA (Tipping and Bishop, 1999a), temporal models).
 - ▶ Model selection through Bayesian treatment of parameters (Bishop, 1999).
 - ▶ Marginalisation of missing data (Tipping and Bishop, 1999b).

Why Probabilistic PCA?

- What is the point in probabilistic methods?
- Could we not just project with regular PCA?
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Note: These same advantages hold for Factor Analysis

Oil and Missing Data

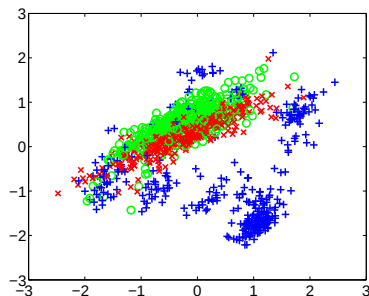
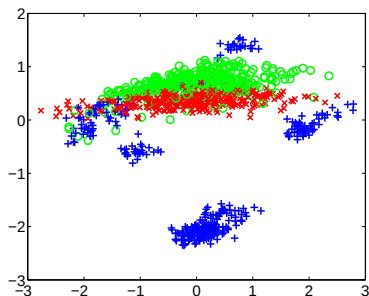


Figure: Projection of the oil data set on to $q = 2$ latent dimensions. *Left:* full data set with no missing data. *Right:* data set with 10% values missing at random.

Oil and Missing Data

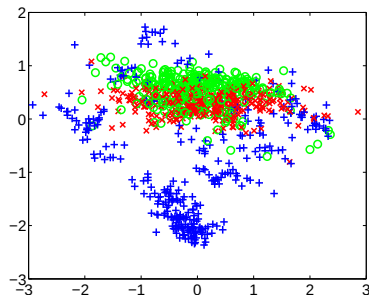
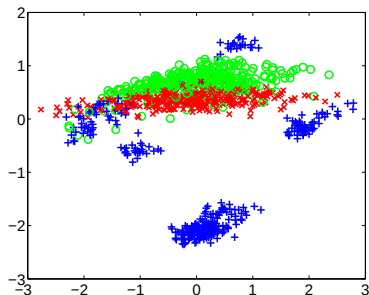


Figure: Projection of the oil data set on to $q = 2$ latent dimensions. *Left:* full data set with no missing data. *Right:* data set with 20% values missing at random.

Oil and Missing Data

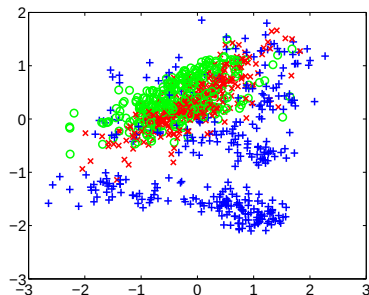
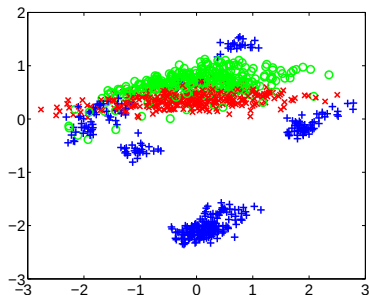


Figure: Projection of the oil data set on to $q = 2$ latent dimensions. *Left:* full data set with no missing data. *Right:* data set with 30% values missing at random.

Oil and Missing Data

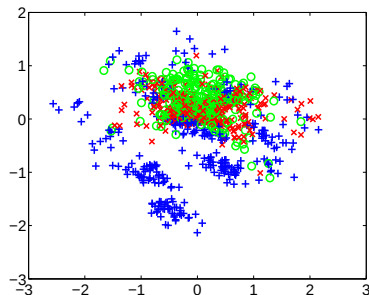
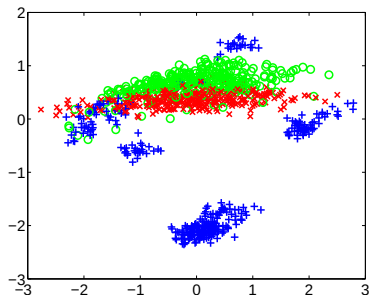


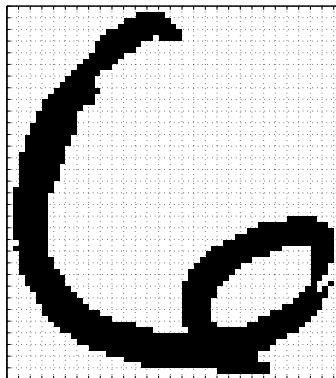
Figure: Projection of the oil data set on to $q = 2$ latent dimensions. *Left:* full data set with no missing data. *Right:* data set with 50% values missing at random.

- Distributions can behave very non-intuitively in high dimensions.
- Fortunately, most data is not really high dimensional.
- Probabilistic PCA exploits linear low dimensional structure in the data.
 - ▶ Probabilistic interpretation brings with it many advantages: extensibility, Bayesian approaches, missing data.
- We will now motivate the need for *non linear* dimensionality reduction.

- 1 Thinking in High Dimensions
- 2 Probabilistic Dimensionality Reduction
- 3 Spectral Methods and Probabilistic Approaches**
- 4 Density Networks and GTM
- 5 Dual Probabilistic PCA and GP-LVM
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USPS Data Set Handwritten Digit

- 3648 Dimensions
 - ▶ 64 rows by 57 columns
 - ▶ Space contains more than just this digit.
 - ▶ Even if we sample every nanosecond from now until the end of the universe, you won't see the original six!



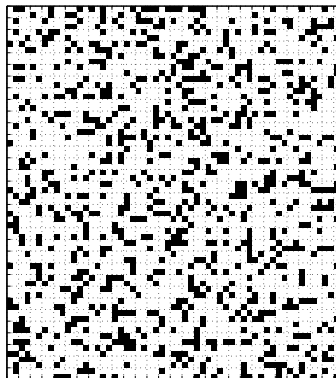
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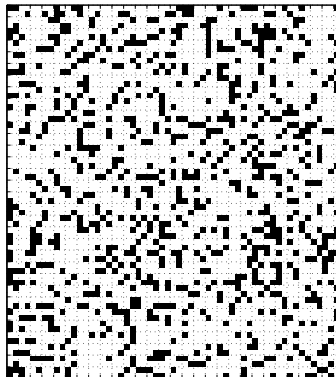
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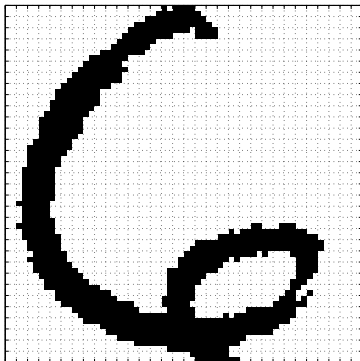


USPS Data Set Handwritten Digit

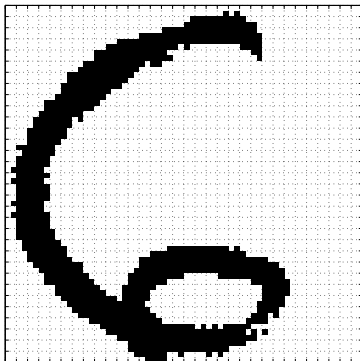
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Rotate a 'Prototype'

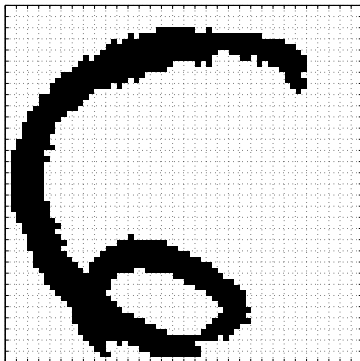


Rotate a 'Prototype'

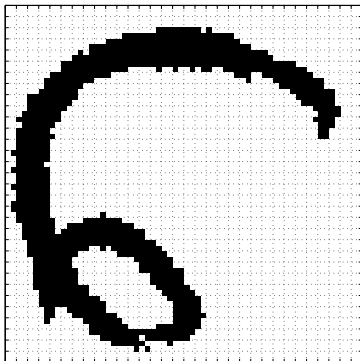


Simple Model of Digit

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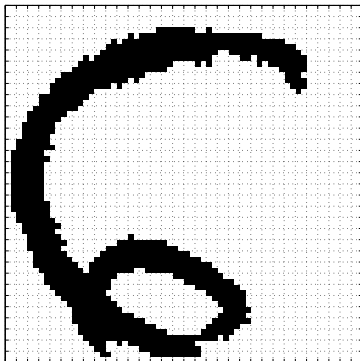


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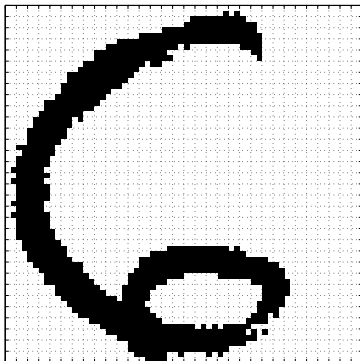


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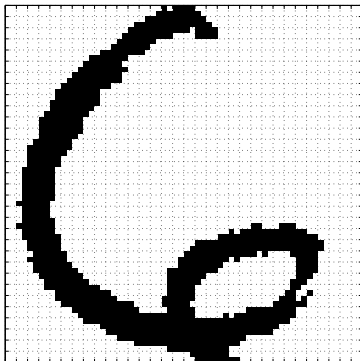
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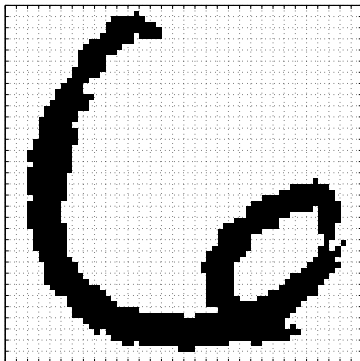
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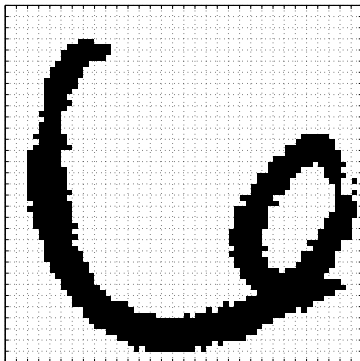
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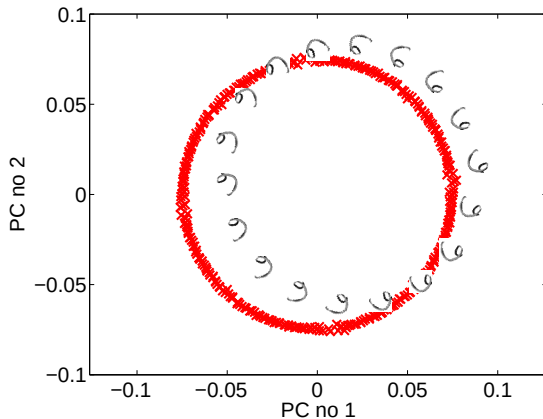


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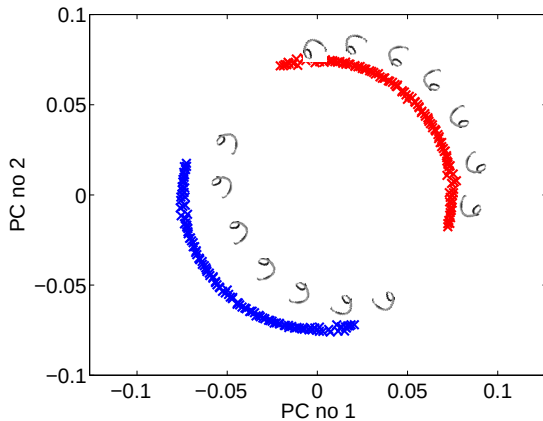


```
demDigitsManifold[1 2], 'all')
```

```
demDigitsManifold[1 2], 'all')
```



```
demDigitsManifold([1 2], 'sixnine')
```



Pure Rotation is too Simple

- In practice the data may undergo several distortions.
 - ▶ e.g. digits undergo 'thinning', translation and rotation.
- For data with 'structure':
 - ▶ we expect fewer distortions than dimensions;
 - ▶ we therefore expect the data to live on a lower dimensional manifold.
- Conclusion: deal with high dimensional data by looking for lower dimensional non-linear embedding.

Spectral Approaches

- Classical Multidimensional Scaling (MDS) (Mardia et al., 1979).
 - ▶ Uses eigenvectors of similarity matrix.
- Kernel PCA (Schölkopf et al., 1998)
 - ▶ Provides a representation and a mapping — representation is high dimensional though!
 - ▶ Mapping is implied through the use of a kernel function as a similarity matrix.
- Isomap (Tenenbaum et al., 2000) is MDS with a particular proximity measure.
 - ▶ Approximate distances measures along the manifold.
 - ▶ Compute neighborhood and compute shortest distance in graph.
 - ▶ Use classical MDS on that distance matrix.

Non Probabilistic Existing Methods II

- Locally Linear Embedding (Roweis and Saul, 2000).
 - ▶ Looks to preserve locally linear relationships in a low dimensional space.
 - ▶ Compute neighborhood and point find reduced dimensional relationships that preserve local linearity.
- Laplacian Eigenmaps (Belkin and Niyogi, 2003).
 - ▶ Uses spectral graph theory and information geometric arguments to form embedding.
 - ▶ Compute neighborhood, graph Laplacian and seek 2nd lowest eigenvector.
- Maximum Variance Unfolding (Weinberger et al., 2004).
 - ▶ Compute neighborhood, constrain local distances to be preserved.
 - ▶ Maximise the variance in latent space.

Iterative Methods

- Multidimensional Scaling (MDS)
 - ▶ Iterative optimisation of a stress function (Kruskal, 1964).
- Sammon Mappings (Sammon, 1969).
 - ▶ Strictly speaking not a mapping — similar to iterative MDS.
- NeuroScale (Lowe and Tipping, 1997)
 - ▶ Augmentation of iterative MDS methods with a mapping.

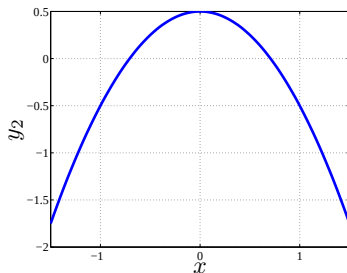
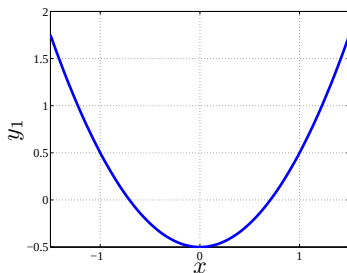
Local Distance Preservation

- Most of the above dimensional reduction techniques preserve local distances.
 - ▶ Probabilistic Approaches do not.
- Probabilistic approaches map smoothly from latent to data space.
 - ▶ Points close in latent space are close in data space.
 - ▶ This does not imply points close in data space are close in latent space.
- Spectral approaches map smoothly from data to latent space.
 - ▶ Points close in data space are close in latent space.
 - ▶ This does not imply points close in latent space are close in data space.

Forward Mapping

- Mapping from 1-D latent space to 2-D data space.

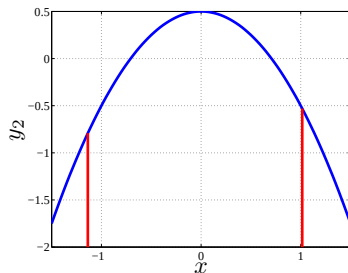
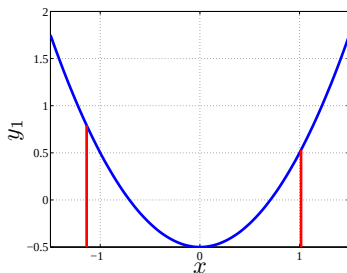
$$y_1 = x^2 - 0.5, \quad y_2 = -x^2 + 0.5$$



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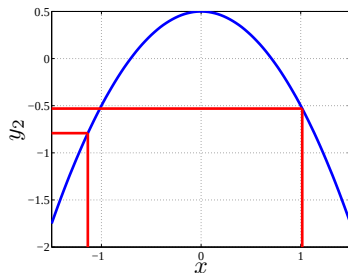
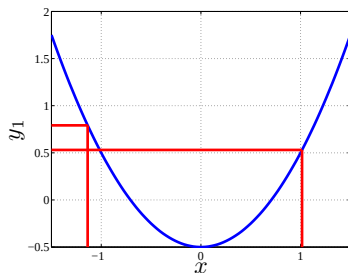
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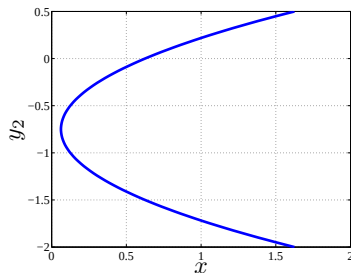
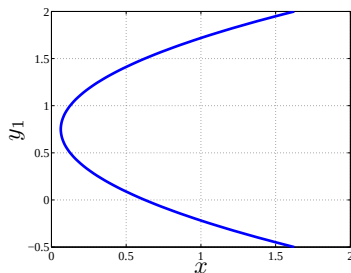
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Backward Mapping

- Mapping from 2-D data space to 1-D latent.

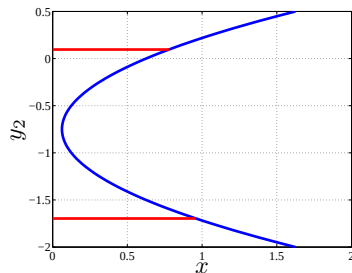
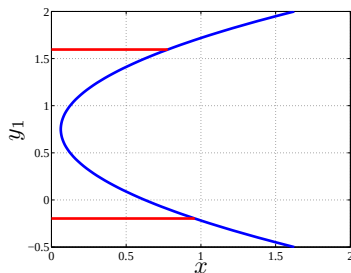
$$x = 0.5 (y_1^2 + y_2^2 + 1)$$



Backward Mapping

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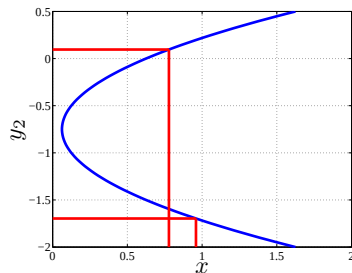
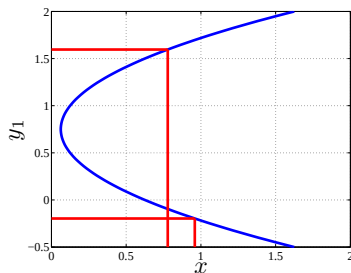
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Backward Mapping

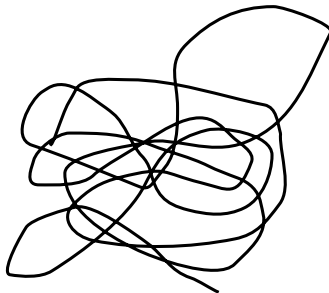
- Mapping from 2-D data space to 1-D latent.

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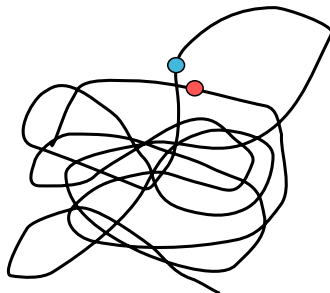
Tangled String

- Sometimes local distance preservation in data space is wrong.
- The pink and blue ball should be separated.
- But the assumption makes the problem simpler (for spectral methods it is convex).



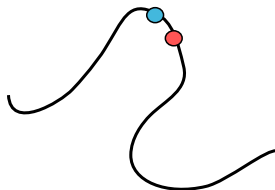
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Good

- Unique optimum.

But

- Non trivial for dealing with missing data.
- Difficult to extend (e.g. temporal data) in a principled way.

- We have motivated the need for non-linear dimensionality reduction.
- Spectral approaches can achieve this, but they don't lead to probabilistic models.
- We are looking for a probabilistic approach to encoding the mapping.
- Next we will see how point based representations of the latent space can be used to achieve this.

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Non Linear Probabilistic Methods I

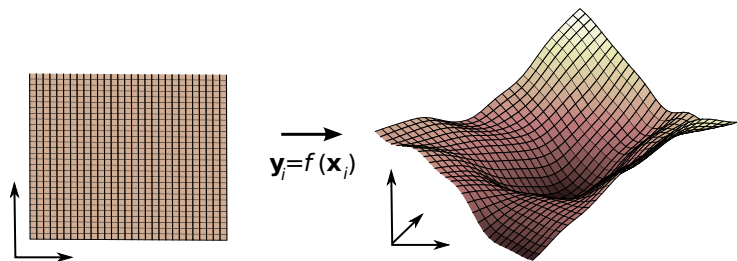


Figure: Mapping a two dimensional plane to a higher dimensional space in a non-linear way.

Difficulty for Probabilistic Approaches

- Propagate a probability distribution through a non-linear mapping.
- Normalisation of distribution becomes intractable.

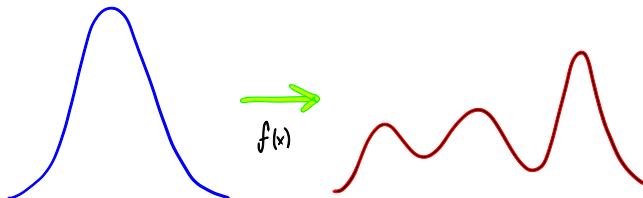


Figure: Gaussian distribution propagated through a non-linear mapping.

Sampling Approach I

- Proposed as Density Networks (MacKay, 1995)
- Likelihood is a Gaussian with non-linear mapping from latent space to data space for the mean

$$p(\mathbf{Y}|\mathbf{X}) = \prod_{i=1}^N \prod_{j=1}^D \mathcal{N}(y_{i,j} | f_j(\mathbf{x}_{i,:}; \boldsymbol{\theta}), \sigma^2)$$

$$p(\mathbf{X}) = \mathcal{N}(\mathbf{x}_{i,:} | \mathbf{0}, \mathbf{I})$$

- Take the mapping to be e.g. a multi-layer perceptron.
- Key idea: share same samples for all data points $\hat{\mathbf{X}}_n = \hat{\mathbf{X}} = \{\hat{\mathbf{x}}_{k,:}\}_{k=1}^M$.
- Saves computation — compute the mapping M times instead of MN

Mapping of Points

- Mapping points to higher dimensions is easy.

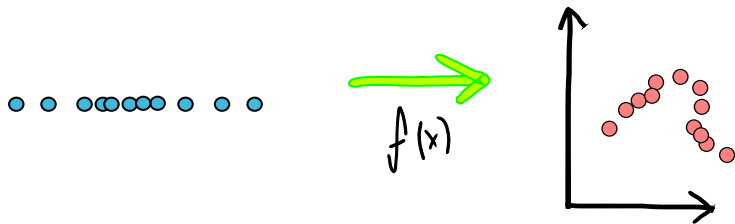


Figure: One dimensional Gaussian mapped to two dimensions.

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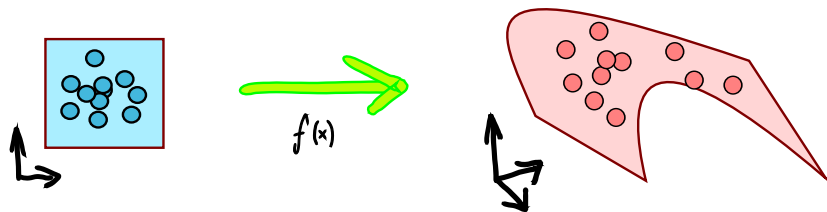


Figure: Two dimensional Gaussian mapped to three dimensions.

Sample approximation to log likelihood:

$$\log p(\mathbf{Y}|\boldsymbol{\theta}) = \sum_{i=1}^N \log \frac{1}{M} \sum_{k=1}^M p(\mathbf{y}_{i,:}|\boldsymbol{\theta}, \bar{\hat{\mathbf{x}}}_{k,:})$$

so we have

$$\frac{d}{d\boldsymbol{\theta}} \log p(\mathbf{y}_{i,:}|\boldsymbol{\theta}) = \sum_{k=1}^M \frac{p(\mathbf{y}_{i,:}|\boldsymbol{\theta}, \hat{\mathbf{x}}_{k,:})}{\sum_{m=1}^M p(\mathbf{y}_{i,:}|\boldsymbol{\theta}, \hat{\mathbf{x}}_{m,:})} \frac{d}{d\boldsymbol{\theta}} \log p(\mathbf{y}_{i,:}|\boldsymbol{\theta}, \hat{\mathbf{x}}_{k,:})$$

$$\frac{d}{d\boldsymbol{\theta}} \log p(\mathbf{y}_{i,:}|\boldsymbol{\theta}) = \sum_{k=1}^M \hat{\pi}_{i,k} \frac{d}{d\boldsymbol{\theta}} \log p(\mathbf{y}_{i,:}|\boldsymbol{\theta}, \hat{\mathbf{x}}_{k,:})$$

Note: $\hat{\pi}_{i,k}$ look a bit like the posterior over component k for data point i .

- Use gradient based optimisation to find the mapping.

demOilDnet4

Figure: Oil data visualised with a density network using an MLP network with points in the sample. Nearest neighbour errors: .

demOilDnet4

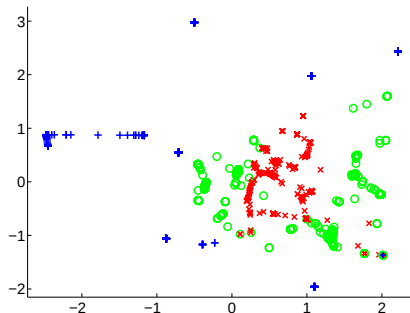


Figure: Oil data visualised with a density network using an MLP network with 100 points in the sample. Nearest neighbour errors: 22.

demOilDnet5

Figure: Oil data visualised with a density network using an MLP network with points in the sample. Nearest neighbour errors: .

demOilDnet5

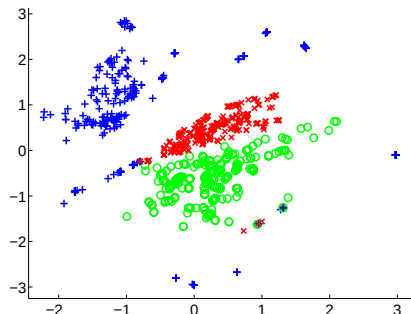


Figure: Oil data visualised with a density network using an MLP network with 400 points in the sample. Nearest neighbour errors: 16.

Generative Topographic Mapping

- Generative Topographic Mapping (GTM) (Bishop et al., 1998a)
- Key idea: Lay points out on a *grid*.
 - ▶ Constrained mixture of Gaussians.

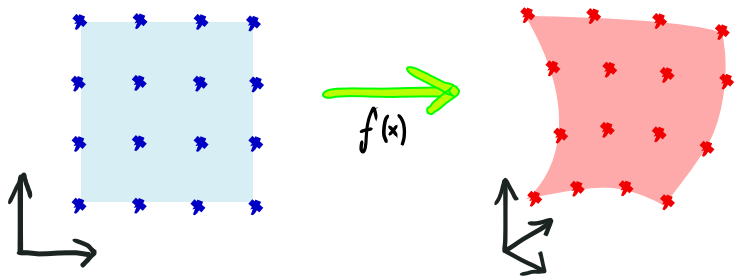


Figure: One dimensional Gaussian mapped to two dimensions.

- Prior distribution is a mixture model in a latent space.

$$p(\mathbf{X}) = \prod_{i=1}^N p(\mathbf{x}_{i,:})$$

$$p(\mathbf{x}_{i,:}) = \frac{1}{M} \sum_{k=1}^M \delta(\mathbf{x}_{i,:} - \hat{\mathbf{x}}_{k,:})$$

- The $\hat{\mathbf{x}}_{k,:}$ are laid out on a regular grid.

- Likelihood is a Gaussian with non-linear mapping from latent space to data space for the mean

$$p(\mathbf{Y}|\mathbf{X}, \theta) = \prod_{i=1}^N \prod_{j=1}^D \mathcal{N}(y_{i,j} | f_j(\mathbf{x}_{i,:}; \mathbf{W}, l), \sigma^2)$$

In the original paper (Bishop et al., 1998b) an RBF network was suggested,

- In the E-step, posterior distribution over k is given by

$$\hat{\pi}_{i,k} = \frac{\prod_{j=1}^D \mathcal{N}(y_{i,j} | f_j(\hat{\mathbf{x}}_k; \mathbf{W}, l), \sigma^2)}{\sum_{m=1}^M \prod_{j=1}^D \mathcal{N}(y_{i,j} | f_j(\hat{\mathbf{x}}_m; \mathbf{W}, l), \sigma^2)}$$

sometimes called the “responsibility of component k for data point i ”.

Likelihood Optimisation

- We then maximise the lower bound on the log likelihood,

$$\log p(\mathbf{y}_{i,:}|\boldsymbol{\theta}) \geq \langle \log p(\mathbf{y}_{i,:}, \hat{\mathbf{x}}_{k,:}|\boldsymbol{\theta}) \rangle_{q(k)} - \langle \log q(k) \rangle_{q(k)},$$

- Free energy part of bound

$$\langle \log p(\mathbf{y}_{i,:}, \hat{\mathbf{x}}_{k,:}|\boldsymbol{\theta}) \rangle = \sum_{k=1}^M \hat{\pi}_{i,k} \log p(\mathbf{y}_{i,:}|\hat{\mathbf{x}}_{k,:}, \boldsymbol{\theta}) + \text{const}$$

- When optimising parameters in EM, we ignore dependence of $\hat{\pi}_{i,k}$ on parameters. So we have

$$\frac{d}{d\boldsymbol{\theta}} \langle \log p(\mathbf{y}_{i,:}, \hat{\mathbf{x}}_{k,:}|\boldsymbol{\theta}) \rangle = \sum_{k=1}^M \hat{\pi}_{i,k} \frac{d}{d\boldsymbol{\theta}} \log p(\mathbf{y}_{i,:}|\hat{\mathbf{x}}_{k,:}, \boldsymbol{\theta})$$

which is very similar to density network result!

- Interpretation of posterior is slightly different.

demOilDnet1

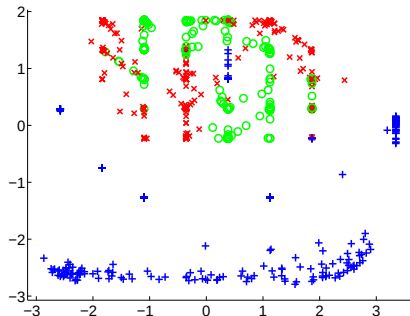


Figure: Oil data visualised with the GTM using an RBF network with 10×10 points in the grid. Nearest neighbour errors: 74.

demOilDnet2

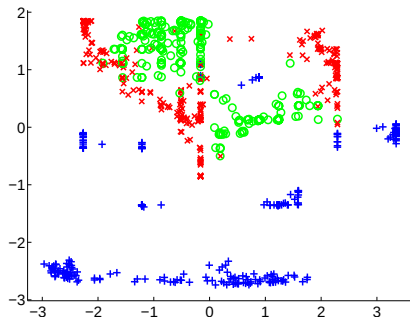


Figure: Oil data visualised with the GTM using an RBF network with 20×20 points in the grid. Nearest neighbour errors: 44.

demOilDnet3

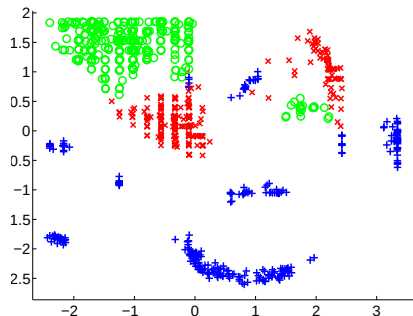


Figure: Oil data visualised with the GTM using an RBF network with 30×30 points in the grid. Nearest neighbour errors: 11.

demStickDnet1

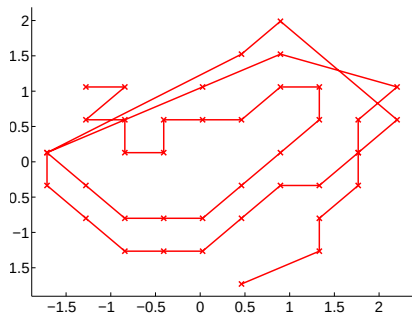


Figure: Oil data visualised with the GTM using an RBF network with 10×10 points in the grid.

demStickDnet2

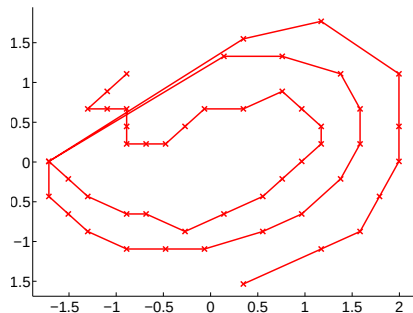


Figure: Oil data visualised with the GTM using an RBF network with 20×20 points in the grid.

demStickDnet3

Figure: Oil data visualised with the GTM using an RBF network with 30×30 points in the grid.

Bubblewrap Effect

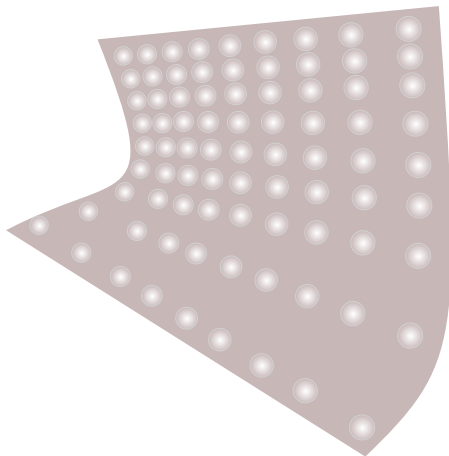
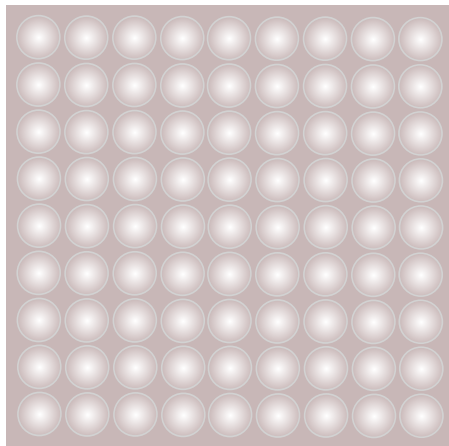


Figure: The manifold is more like bubblewrap than a piece of paper.

Effect of Separated Means

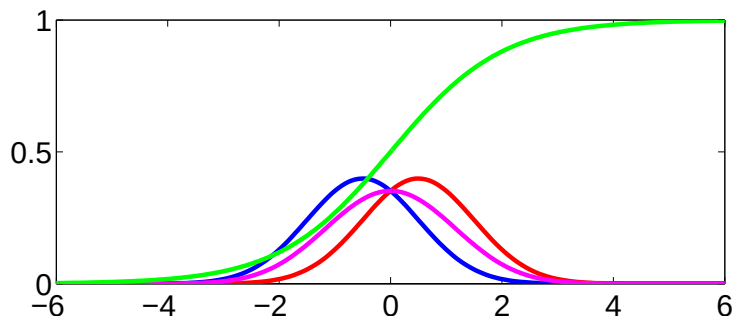


Figure: As Gaussians become further apart the posterior probability becomes more abrupt. 1 standard deviations apart.

Effect of Separated Means

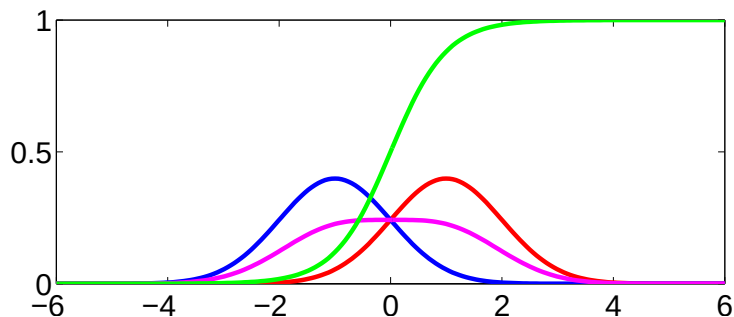


Figure: As Gaussians become further apart the posterior probability becomes more abrupt. 2 standard deviations apart.

Effect of Separated Means

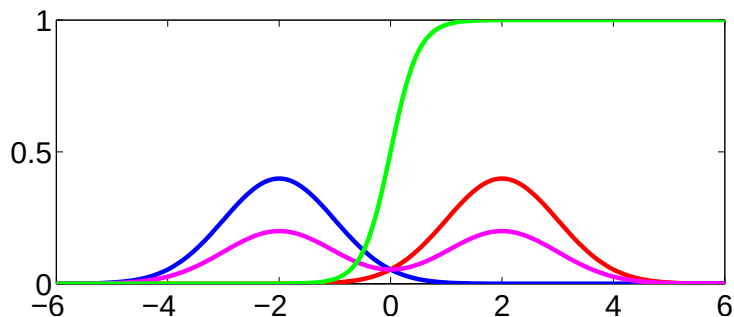


Figure: As Gaussians become further apart the posterior probability becomes more abrupt. 4 standard deviations apart.

Effect of Separated Means

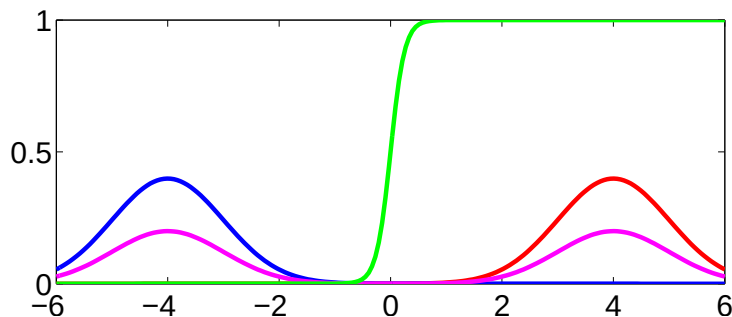


Figure: As Gaussians become further apart the posterior probability becomes more abrupt. 8 standard deviations apart.

Effect of Separated Means

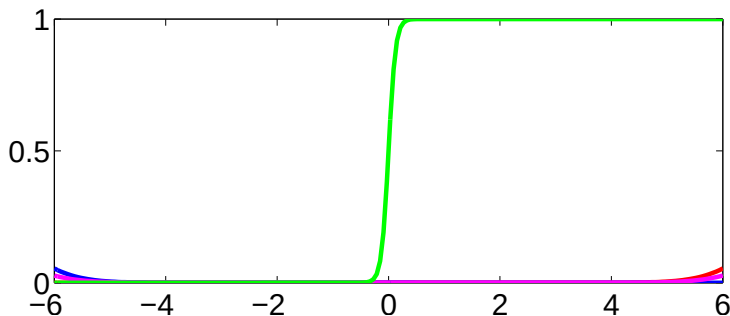


Figure: As Gaussians become further apart the posterior probability becomes more abrupt. 16 standard deviations apart.

Equivalence of GTM and Density Networks

- GTM and Density Networks have the same origin. (Bishop et al., 1996; MacKay, 1995).
- In original Density Networks paper MacKay suggested Importance Sampling (MacKay, 1995).
- Early work on GTM also used importance sampling.
- Main innovation in GTM was to lay points out on a grid (inspired by Self Organizing Maps (Kohonen, 2001)).

Non Linear Factor Analysis

- Variational approach to dimensionality reduction.
- Combine Gaussian prior over latent space with neural network (Honkela and Valpola, 2005)
- Assume variational prior separates.
- Optimise with respect to variational distributions.

- We have explored two point based approaches to dimensionality reduction.
- Approaches seem to generalise well even when dimensions of data is greater than number of points.
- Both approaches are difficult to extend to higher dimensional latent spaces
 - ▶ number of samples/centres required increases exponentially with dimension.
- Next we will explore a different probabilistic interpretation of PCA and extend that to non-linear models.

Outline

- 1 Thinking in High Dimensions
- 2 Probabilistic Dimensionality Reduction
- 3 Spectral Methods and Probabilistic Approaches
- 4 Density Networks and GTM
- 5 Dual Probabilistic PCA and GP-LVM**
- 6 Applications
- 7 Conclusions

Probabilistic PCA

- We have seen that PCA has a probabilistic interpretation (Tipping and Bishop, 1999b).
- It is difficult to 'non-linearise' directly.
- GTM and Density Networks are an attempt to do so.

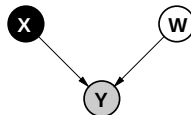
Dual Probabilistic PCA

- There is an alternative probabilistic interpretation of PCA (Lawrence, 2005).
- This interpretation can be made non-linear.
- The result is non-linear probabilistic PCA.

Dual Probabilistic PCA

- Define *linear-Gaussian relationship* between latent variables and data.

- ▶ **Novel** Latent variable approach:
- ▶ Define Gaussian prior over *parameters*, $\mathbf{W}$.
- ▶ Integrate out *parameters*.

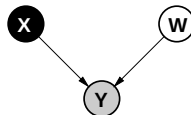


$$p(\mathbf{Y}|\mathbf{X}, \mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{W}\mathbf{x}_{i,:}, \sigma^2 \mathbf{I})$$

Dual Probabilistic PCA

- Define *linear-Gaussian relationship* between latent variables and data.

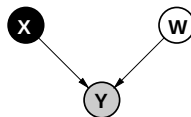
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Dual Probabilistic PCA

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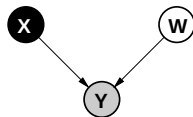


$$p(\mathbf{Y}|\mathbf{X}, \mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{W}\mathbf{x}_{i,:}, \sigma^2 \mathbf{I})$$

$$p(\mathbf{W}) = \prod_{i=1}^D \mathcal{N}(\mathbf{w}_{i,:} | \mathbf{0}, \mathbf{I})$$

Dual Probabilistic PCA

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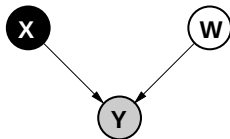


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$$p(\mathbf{Y}|\mathbf{X}) = \prod_{j=1}^D \mathcal{N}(\mathbf{y}_{:,j} | \mathbf{0}, \mathbf{X}\mathbf{X}^T + \sigma^2 \mathbf{I})$$

Dual Probabilistic PCA Max. Likelihood Soln (Lawrence, 2004)



$$p(\mathbf{Y}|\mathbf{X}) = \prod_{j=1}^D \mathcal{N}(\mathbf{y}_{:,j} | \mathbf{0}, \mathbf{X}\mathbf{X}^T + \sigma^2 \mathbf{I})$$

Dual Probabilistic PCA Max. Likelihood Soln (Lawrence, 2004)

$$p(\mathbf{Y}|\mathbf{X}) = \prod_{j=1}^D \mathcal{N}(\mathbf{y}_{:,j} | \mathbf{0}, \mathbf{K}), \quad \mathbf{K} = \mathbf{X}\mathbf{X}^T + \sigma^2 \mathbf{I}$$

$$\log p(\mathbf{Y}|\mathbf{X}) = -\frac{D}{2} \log |\mathbf{K}| - \frac{1}{2} \text{tr}(\mathbf{K}^{-1} \mathbf{Y}\mathbf{Y}^T) + \text{const.}$$

If $\mathbf{U}'_q$ are first q principal eigenvectors of $D^{-1} \mathbf{Y}\mathbf{Y}^T$ and the corresponding eigenvalues are Λ_q ,

$$\mathbf{X} = \mathbf{U}'_q \mathbf{L} \mathbf{R}^T, \quad \mathbf{L} = (\Lambda_q - \sigma^2 \mathbf{I})^{\frac{1}{2}}$$

where $\mathbf{R}$ is an arbitrary rotation matrix.

Probabilistic PCA Max. Likelihood Soln (Tipping and Bishop, 1999b)

$$p(\mathbf{Y}|\mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{0}, \mathbf{C}), \quad \mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}$$

$$\log p(\mathbf{Y}|\mathbf{W}) = -\frac{N}{2} \log |\mathbf{C}| - \frac{1}{2} \text{tr}(\mathbf{C}^{-1} \mathbf{Y}^T \mathbf{Y}) + \text{const.}$$

If $\mathbf{U}_q$ are first q principal eigenvectors of $N^{-1} \mathbf{Y}^T \mathbf{Y}$ and the corresponding eigenvalues are Λ_q ,

$$\mathbf{W} = \mathbf{U}_q \mathbf{L} \mathbf{R}^T, \quad \mathbf{L} = (\Lambda_q - \sigma^2 \mathbf{I})^{\frac{1}{2}}$$

where $\mathbf{R}$ is an arbitrary rotation matrix.

The Eigenvalue Problems are equivalent

- Solution for Probabilistic PCA (solves for the mapping)

$$\mathbf{Y}^T \mathbf{Y} \mathbf{U}_q = \mathbf{U}_q \Lambda_q \quad \mathbf{W} = \mathbf{U}_q \mathbf{L} \mathbf{V}^T$$

- Solution for Dual Probabilistic PCA (solves for the latent positions)

$$\mathbf{Y} \mathbf{Y}^T \mathbf{U}'_q = \mathbf{U}'_q \Lambda_q \quad \mathbf{X} = \mathbf{U}'_q \mathbf{L} \mathbf{V}^T$$

- Equivalence is from

$$\mathbf{U}_q = \mathbf{Y}^T \mathbf{U}'_q \Lambda_q^{-\frac{1}{2}}$$

Prior for Functions

- Probability Distribution over Functions
- Functions are infinite dimensional.
 - ▶ Prior distribution over *instantiations* of the function: finite dimensional objects.
 - ▶ Can prove by induction that GP is 'consistent'.
- Mean and Covariance Functions
- Instead of mean and covariance matrix, GP is defined by mean function and covariance function.
 - ▶ Mean function often taken to be zero or constant.
 - ▶ Covariance function must be *positive definite*.
 - ▶ Class of valid covariance functions is the same as the class of *Mercer kernels*.

Zero mean Gaussian Process

- A (zero mean) Gaussian process likelihood is of the form

$$p(\mathbf{y}|\mathbf{X}) = N(\mathbf{y}|\mathbf{0}, \mathbf{K}),$$

where $\mathbf{K}$ is the covariance function or *kernel*.

- The *linear kernel* with noise has the form

$$\mathbf{K} = \mathbf{X}\mathbf{X}^T + \sigma^2\mathbf{I}$$

- Priors over non-linear functions are also possible.
 - ▶ To see what functions look like, we can sample from the prior process.

demCovFuncSample

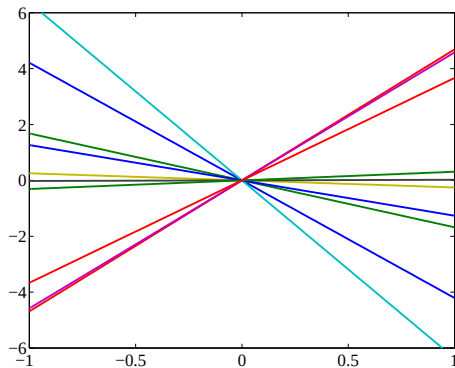


Figure: linear kernel, $\mathbf{K} = \mathbf{X}\mathbf{X}^T$

demCovFuncSample

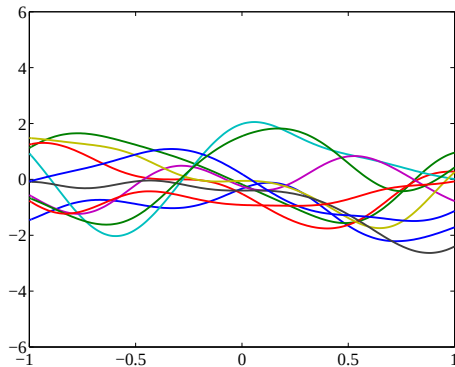


Figure: RBF kernel with $\gamma = 10$, $\alpha = 1$

demCovFuncSample

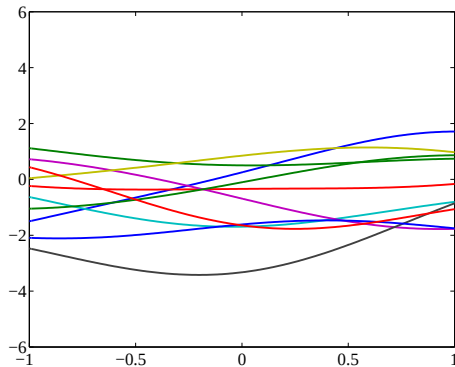


Figure: RBF kernel with $l = 1$, $\alpha = 1$

demCovFuncSample

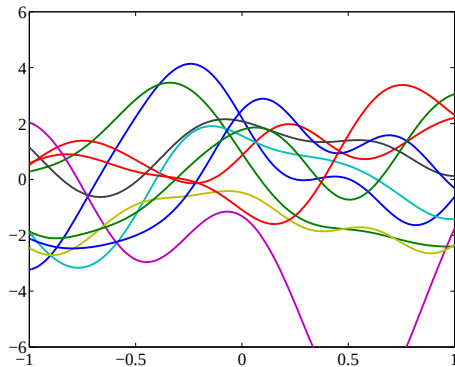


Figure: RBF kernel with $l = 0.3$, $\alpha = 4$

demCovFuncSample

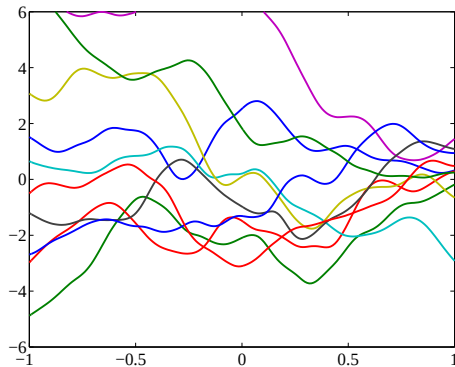


Figure: MLP kernel with $\alpha = 8$, $w = 100$ and $b = 100$

demCovFuncSample

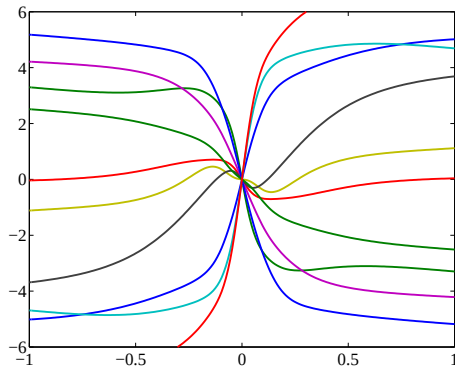


Figure: MLP kernel with $\alpha = 8$, $b = 0$ and $w = 100$

demCovFuncSample

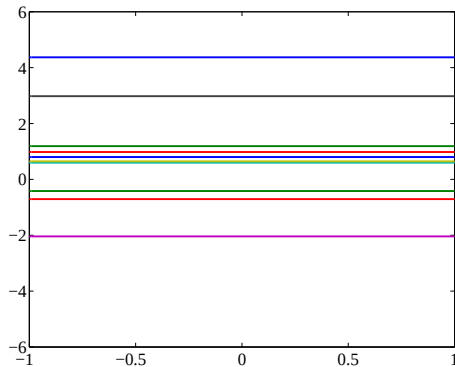


Figure: bias kernel with $\alpha = 1$ and

demCovFuncSample

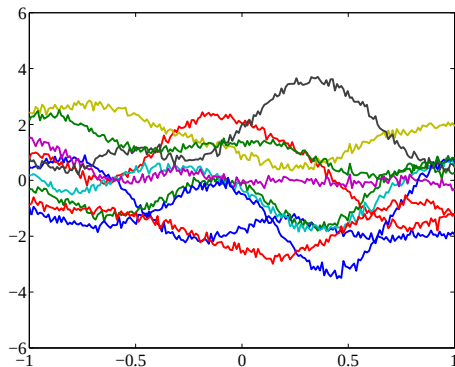


Figure: summed combination of: RBF kernel, $\alpha = 1$, $l = 0.3$; bias kernel, $\alpha = 1$; and white noise kernel, $\beta = 100$

Posterior Distribution over Functions

- Gaussian processes are often used for regression.
- We are given a known inputs $\mathbf{X}$ and targets $\mathbf{Y}$.
- We assume a prior distribution over functions by selecting a kernel.
- Combine the prior with data to get a *posterior* distribution over functions.

Gaussian Process Regression

demRegression

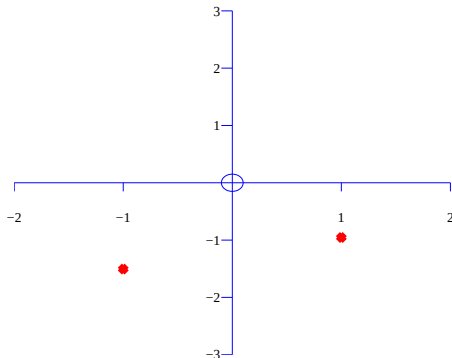


Figure: Examples include WiFi localization, C14 calibration curve.

Gaussian Process Regression

demRegression

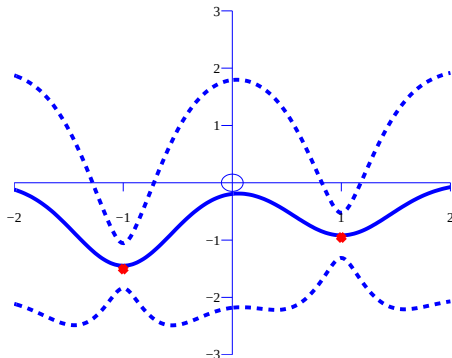


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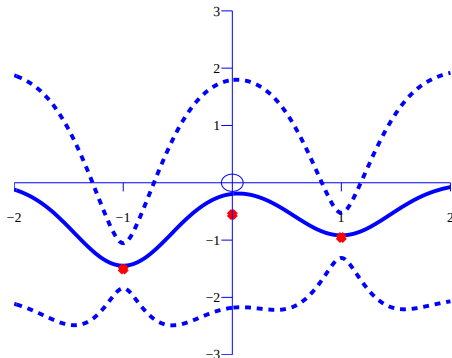


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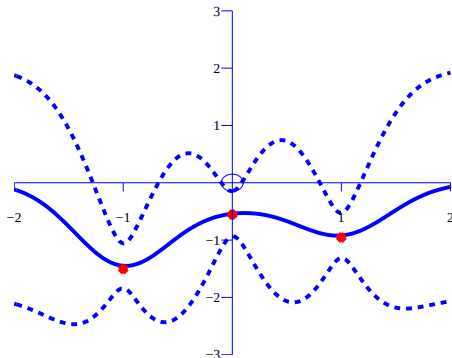


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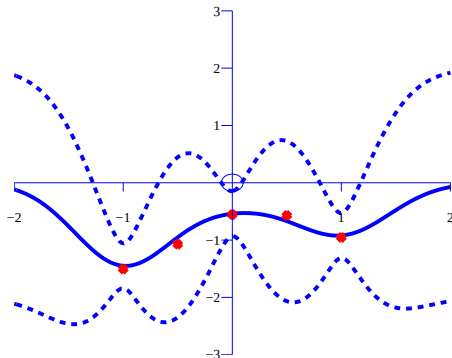


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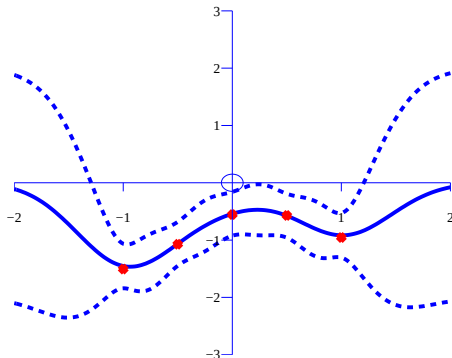


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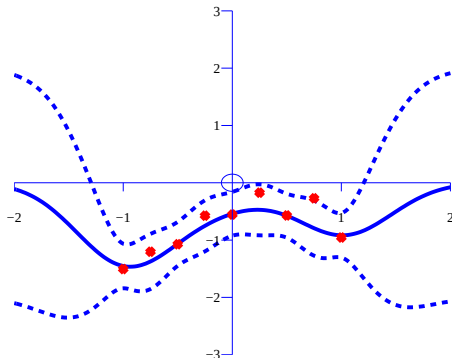


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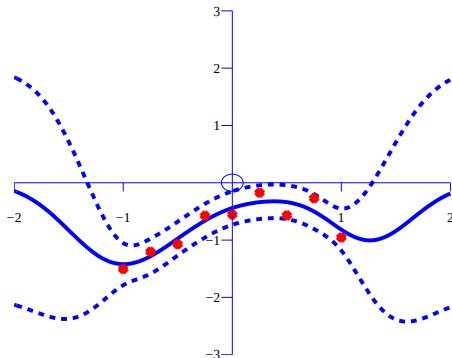
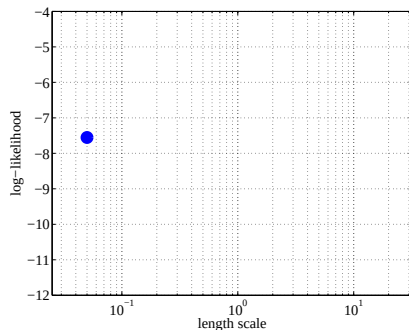
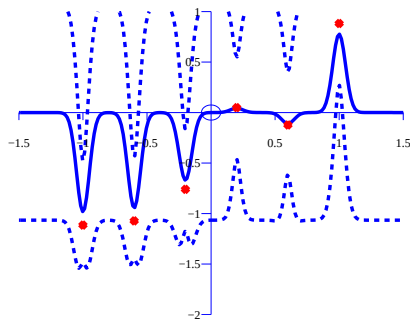


Figure: Examples include WiFi localization, C14 calibration curve.

Learning Kernel Parameters

Can we determine length scales and noise levels from the data?

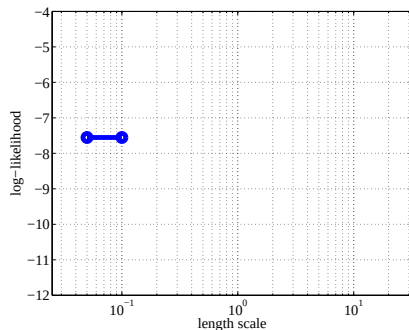
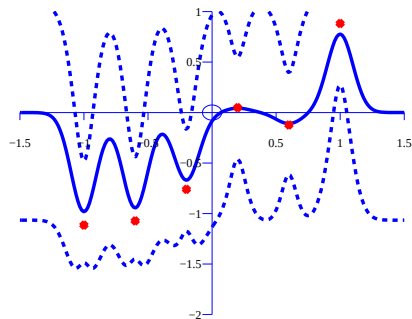
demOptimiseKern



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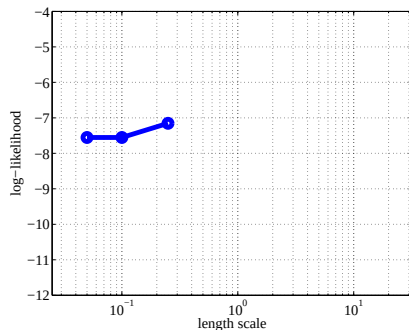
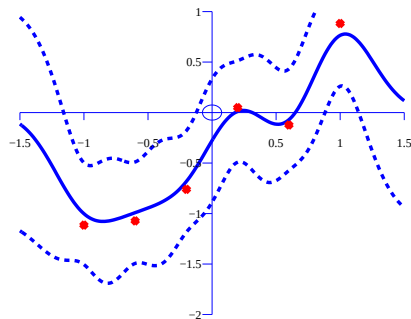
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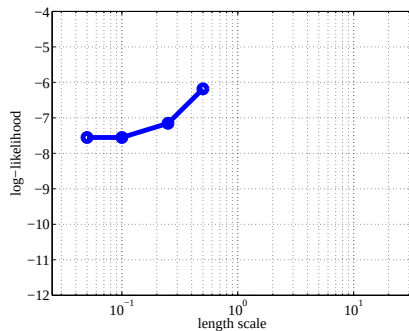
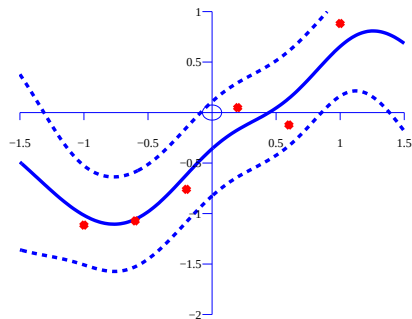
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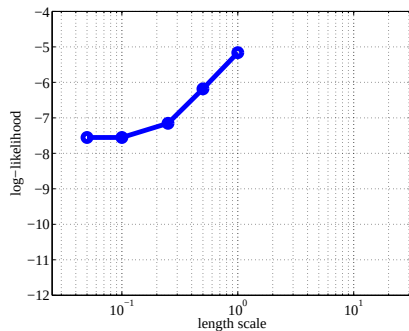
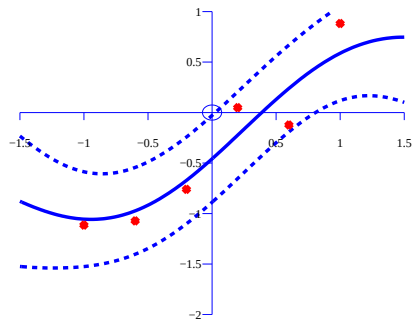
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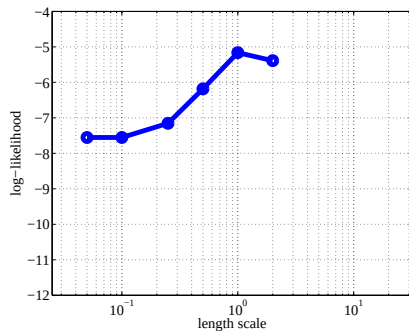
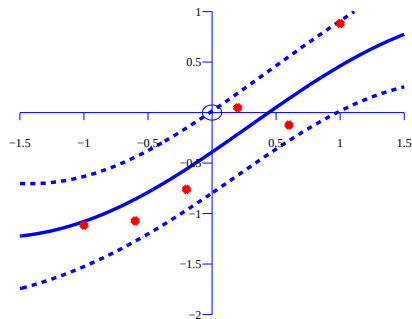
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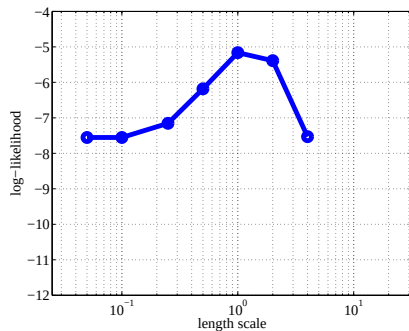
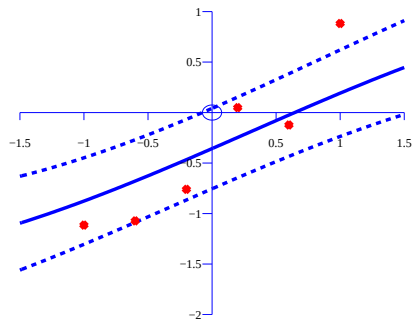
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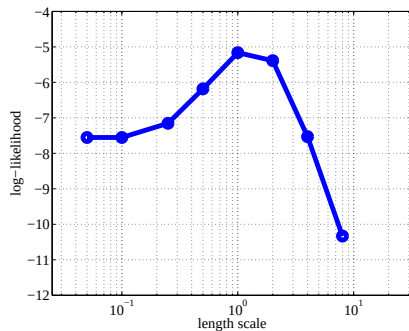
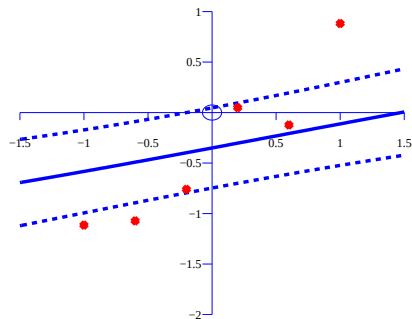
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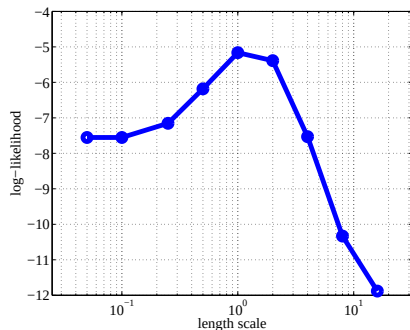
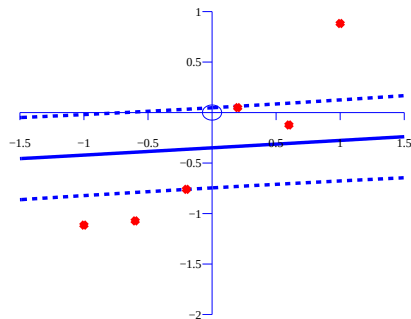
demOptimiseKern



Learning Kernel Parameters

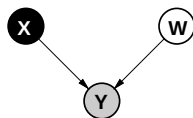
Can we determine length scales and noise levels from the data?

demOptimiseKern



Dual Probabilistic PCA

- Define *linear-Gaussian relationship* between latent variables and data.
- **Novel** Latent variable approach:
 - ▶ Define Gaussian prior over *parameters*, $\mathbf{W}$.
 - ▶ Integrate out *parameters*.



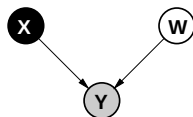
$$p(\mathbf{Y}|\mathbf{X}, \mathbf{W}) = \prod_{i=1}^n N(\mathbf{y}_{i,:} | \mathbf{W}\mathbf{x}_{i,:}, \sigma^2 \mathbf{I})$$

$$p(\mathbf{W}) = \prod_{i=1}^d N(\mathbf{w}_{i,:} | \mathbf{0}, \mathbf{I})$$

$$p(\mathbf{Y}|\mathbf{X}) = \prod_{j=1}^d N(\mathbf{y}_{:,j} | \mathbf{0}, \mathbf{X}\mathbf{X}^T + \sigma^2 \mathbf{I})$$

Dual Probabilistic PCA

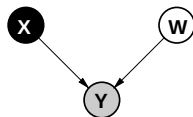
- Inspection of the marginal likelihood shows ...
 - ▶ The covariance matrix is a covariance function.
 - ▶ We recognise it as the 'linear kernel'.



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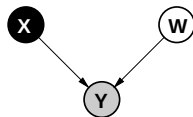


$$p(\mathbf{Y}|\mathbf{X}) = \prod_{j=1}^d N(\mathbf{y}_{:,j} | \mathbf{0}, \mathbf{K})$$

$$\mathbf{K} = \mathbf{X}\mathbf{X}^T + \sigma^2\mathbf{I}$$

Dual Probabilistic PCA

- Inspection of the marginal likelihood shows ...
 - ▶ The covariance matrix is a covariance function.
 - ▶ We recognise it as the 'linear kernel'.



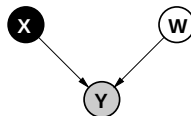
$$p(\mathbf{Y}|\mathbf{X}) = \prod_{j=1}^d N(\mathbf{y}_{:,j} | \mathbf{0}, \mathbf{K})$$

$$\mathbf{K} = \mathbf{X}\mathbf{X}^T + \sigma^2\mathbf{I}$$

This is a product of Gaussian processes
with linear kernels.

Dual Probabilistic PCA

- Inspection of the marginal likelihood shows ...
 - ▶ The covariance matrix is a covariance function.
 - ▶ We recognise it as the 'linear kernel'.



$$p(\mathbf{Y}|\mathbf{X}) = \prod_{j=1}^d \mathcal{N}(\mathbf{y}_{:,j} | \mathbf{0}, \mathbf{K})$$

$$\mathbf{K} = ?$$

Replace linear kernel with non-linear
kernel for non-linear model.

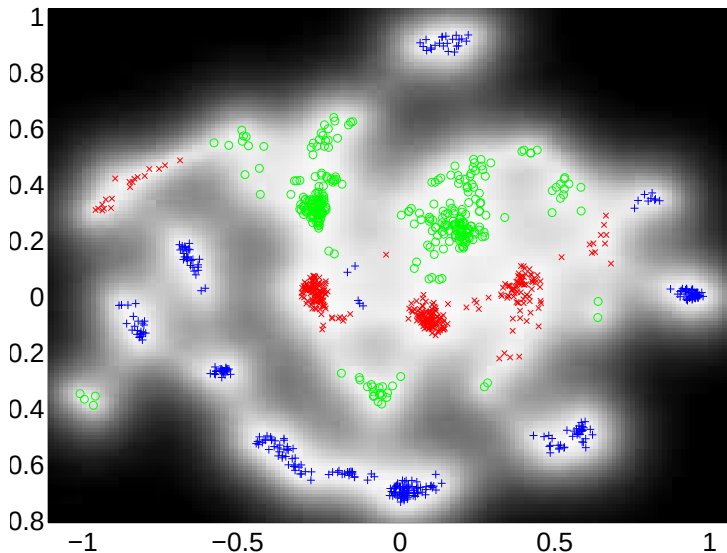
RBF Kernel

- The RBF kernel has the form $k_{i,j} = k(\mathbf{x}_{i,:}, \mathbf{x}_{j,:})$, where

$$k(\mathbf{x}_{i,:}, \mathbf{x}_{j,:}) = \alpha \exp \left(-\frac{(\mathbf{x}_{i,:} - \mathbf{x}_{j,:})^T (\mathbf{x}_{i,:} - \mathbf{x}_{j,:})}{2l^2} \right).$$

- No longer possible to optimise wrt $\mathbf{X}$ via an eigenvalue problem.
- Instead find gradients with respect to $\mathbf{X}$, α , l and σ^2 and optimise using gradient methods.

Oil Data Set



Nearest Neighbour error in X

- Nearest neighbour classification in latent space.

Method	PCA	GTM	GP-LVM
Errors	162	11	1

cf 2 errors in data space.

demStick1

Figure: The latent space for the stick man motion capture data.

demStick1

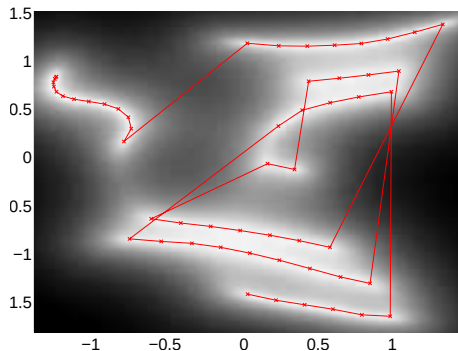
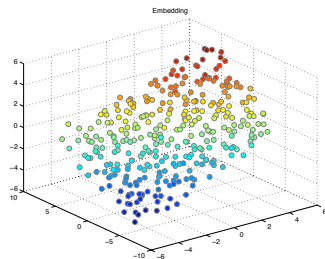


Figure: The latent space for the stick man motion capture data.

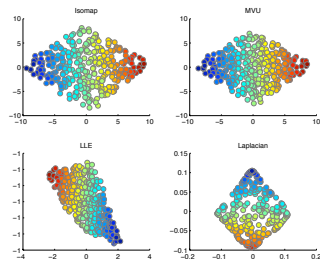
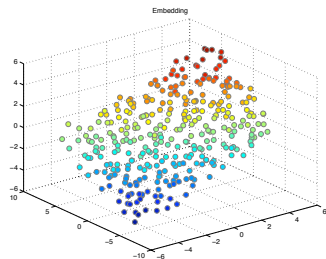
- *Lawrence* - 2003 suggested the use of Spectral algorithms to initialise the latent space $\mathbf{Y}$
- *Harmeling* - 2007 evaluated the use of GP-LVM objective for model selection
 - ▶ Comparisons between **Procrustes** score to ground truth and GP-LVM objective

Model Selection: Results¹



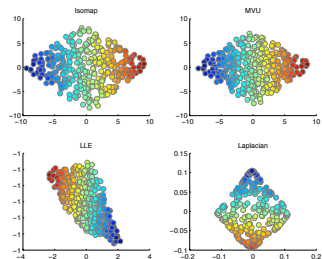
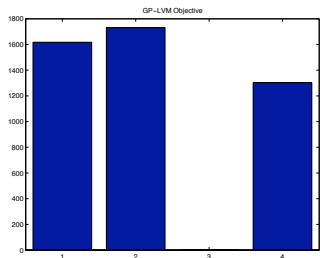
¹Model selection results kindly provided by Carl Henrik Ek.

Model Selection: Results¹



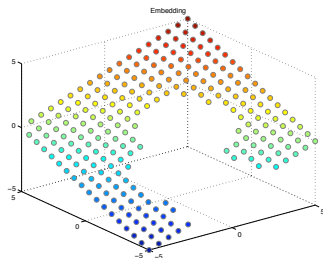
¹Model selection results kindly provided by Carl Henrik Ek.

Model Selection: Results¹



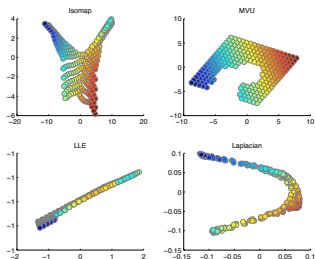
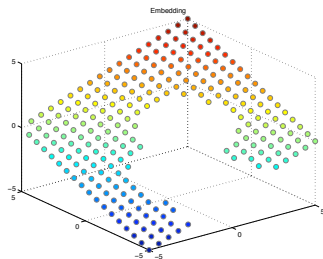
¹Model selection results kindly provided by Carl Henrik Ek.

Model Selection: Results¹



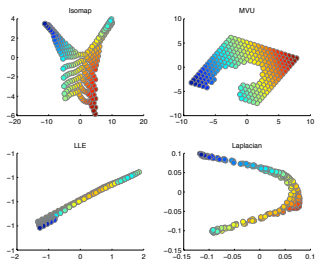
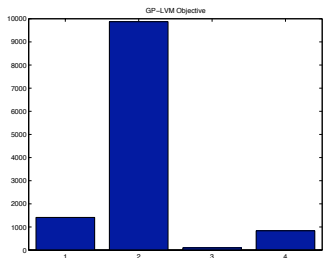
¹Model selection results kindly provided by Carl Henrik Ek.

Model Selection: Results¹



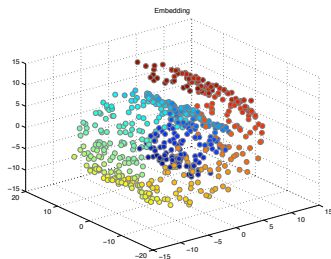
¹Model selection results kindly provided by Carl Henrik Ek.

Model Selection: Results¹



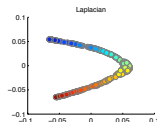
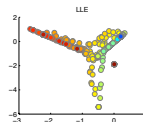
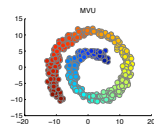
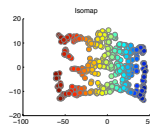
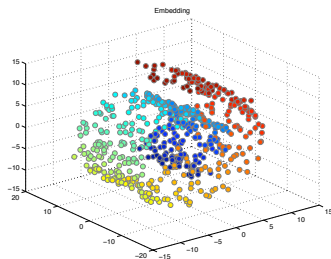
¹Model selection results kindly provided by Carl Henrik Ek.

Model Selection: Results¹



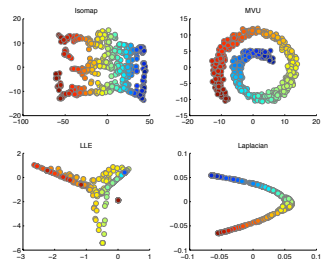
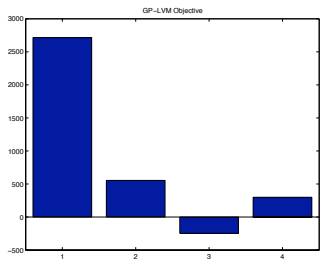
¹Model selection results kindly provided by Carl Henrik Ek.

Model Selection: Results¹



¹Model selection results kindly provided by Carl Henrik Ek.

Model Selection: Results¹



¹Model selection results kindly provided by Carl Henrik Ek.

- We introduced a dual probabilistic interpretation of PCA.
- It was straightforward to non-linearise it using Gaussian processes.
- Result is a non-linear probabilistic PCA.
- Optimise latent variables rather than integrate them out.
- Next: some examples of the model used in applications.

- Style based inverse kinematics (Grochow et al., 2004).
- Prior distributions for tracking (Urtasun et al., 2006, 2005).
- Assisted drawing (Baxter and ichi Anjyo, 2006).

- Initialisation [▶ Details](#)
- Local distance preservation [▶ Details](#)
- Dynamical models [▶ Details](#)
- Hierarchical models [▶ Details](#)
- Linear Back Constraints [▶ Details](#)
- Oil Data Comparison [▶ Details](#)
- Vowel data [▶ Details](#)
- WiFi SLAM [▶ Details](#)
- Non-Gaussian data [▶ Details](#)

- We've advocated Dimensionality Reduction as a good way of *probabilistic* modelling in high dimensions.
- Probabilistic techniques map the “correct way” around.
 - ▶ This leads to problems with local minima.
- Probabilistic dimensionality reduction is useful in practice.
- There are still many open problems to be overcome.

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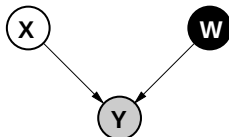
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Probabilistic PCA Max. Likelihood Soln (Tipping and Bishop, 1999b)

- Implications: search high dimensional data by assuming it is implicitly lower dimensional.



$$p(\mathbf{Y}|\mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{0}, \mathbf{W}\mathbf{W}^T + \sigma^2 \mathbf{I})$$

Maximum Likelihood Solution

Probabilistic PCA Max. Likelihood Soln (Tipping and Bishop, 1999b)

- Implications: search high dimensional data by assuming it is implicitly lower dimensional.

$$p(\mathbf{Y}|\mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{0}, \mathbf{W}\mathbf{W}^T + \sigma^2 \mathbf{I})$$

$$p(\mathbf{Y}|\mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{0}, \mathbf{C}), \quad \mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2 \mathbf{I}$$

Gradient of log likelihood

$$\frac{d}{d\mathbf{W}} \log p(\mathbf{Y}|\mathbf{W}) = -\frac{N}{2} \mathbf{C}^{-1} \mathbf{W} + \frac{1}{2} \mathbf{C}^{-1} \mathbf{Y}^T \mathbf{Y} \mathbf{C}^{-1} \mathbf{W}$$

Seek fixed points

$$\mathbf{0} = -\frac{N}{2} \mathbf{C}^{-1} \mathbf{W} + \frac{1}{2} \mathbf{C}^{-1} \mathbf{Y}^T \mathbf{Y} \mathbf{C}^{-1} \mathbf{W}$$

pre-multiply by $2\mathbf{C}$

Maximum Likelihood Solution

Probabilistic PCA Max. Likelihood Soln (Tipping and Bishop, 1999b)

- Implications: search high dimensional data by assuming it is implicitly lower dimensional.

$$p(\mathbf{Y}|\mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{0}, \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I})$$

$$p(\mathbf{Y}|\mathbf{W}) = \prod_{i=1}^N \mathcal{N}(\mathbf{y}_{i,:} | \mathbf{0}, \mathbf{C}), \quad \mathbf{C} = \mathbf{W}\mathbf{W}^T + \sigma^2\mathbf{I}$$

$$\log p(\mathbf{Y}|\mathbf{W}) = -\frac{N}{2} \log |\mathbf{C}| - \frac{1}{2} \text{tr}(\mathbf{C}^{-1} \mathbf{Y}^T \mathbf{Y}) + \text{const.}$$

Gradient of log likelihood

$$\frac{d}{d\mathbf{W}} \log p(\mathbf{Y}|\mathbf{W}) = -\frac{N}{2} \mathbf{C}^{-1} \mathbf{W} + \frac{1}{2} \mathbf{C}^{-1} \mathbf{Y}^T \mathbf{Y} \mathbf{C}^{-1} \mathbf{W}$$

Seek fixed points

$$\mathbf{0} = -\frac{N}{2} \mathbf{C}^{-1} \mathbf{W} + \frac{1}{2} \mathbf{C}^{-1} \mathbf{Y}^T \mathbf{Y} \mathbf{C}^{-1} \mathbf{W}$$

- If $\mathbf{U}_q$ are first q principal eigenvectors of $N^{-1}\mathbf{Y}^T\mathbf{Y}$ and the corresponding eigenvalues are Λ_q ,

$$\mathbf{W} = \mathbf{U}_q \mathbf{L} \mathbf{R}^T, \quad \mathbf{L} = (\Lambda_q - \sigma^2 \mathbf{I})^{\frac{1}{2}}$$

where $\mathbf{R}$ is an arbitrary rotation matrix.

► Return

'Swiss Roll'

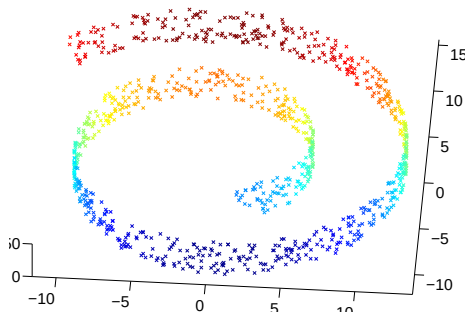


Figure: The 'Swiss Roll' data set is data in three dimensions that is inherently two dimensional.

Quality of solution is Initialisation Dependent

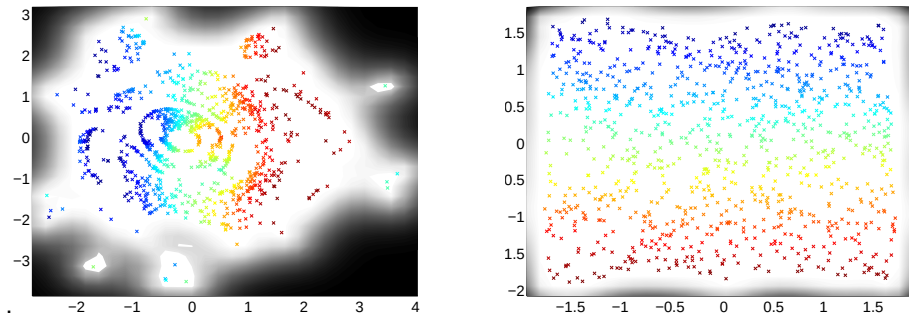


Figure: *Left:* Swiss roll solution initialised by PCA. *Right:* Swiss roll solution initialised by Isomap.

► Return

Multi-Dimensional Scaling with a Mapping

- Lowe and Tipping (1997) made latent positions a function of the data.

$$x_{ij} = f_j(\mathbf{y}_i; \mathbf{w})$$

- ▶ Function was either multi-layer perceptron or a radial basis function network.
- ▶ Their motivation was different from ours:
 - ★ They wanted to add the advantages of a true mapping to multi-dimensional scaling.

Back Constraints

- We can use the same idea to force the GP-LVM to respect local distances. (Lawrence and Quiñonero Candela, 2006)
- By constraining each $\mathbf{x}_i$ to be a 'smooth' mapping from $\mathbf{y}_i$ local distances can be respected.
- This works because in the GP-LVM we maximise wrt latent variables, we don't integrate out.
- Can use any 'smooth' function:
 - 1 Neural network.
 - 2 RBF Network.
 - 3 Kernel based mapping.

Computing Gradients

- GP-LVM normally proceeds by optimising

$$L(\mathbf{X}) = \log p(\mathbf{Y}|\mathbf{X})$$

with respect to $\mathbf{X}$ using $\frac{dL}{d\mathbf{X}}$.

- The back constraints are of the form

$$x_{ij} = f_j(\mathbf{y}_{i,:}; \mathbf{B})$$

where $\mathbf{B}$ are parameters.

- We can compute $\frac{dL}{d\mathbf{B}}$ via chain rule and optimise parameters of mapping.

Motion Capture Results

demStick1 **and** demStick3

Figure: The latent space for the motion capture data with (*right*) and without (*left*) dynamics. The dynamics us a Gaussian process with an RBF kernel.

.

Motion Capture Results

demStick1 and demStick3

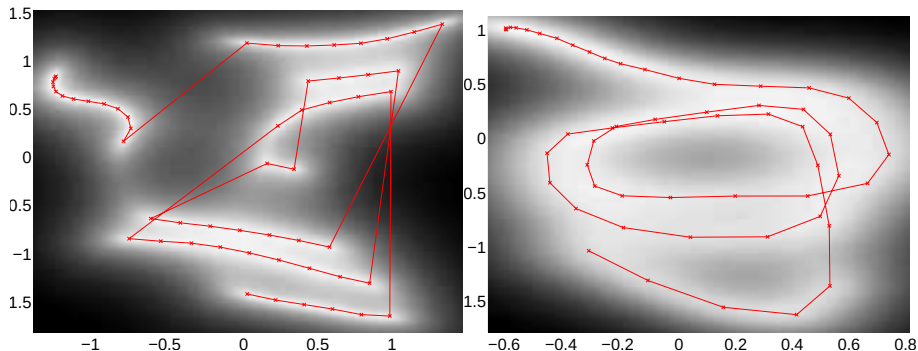
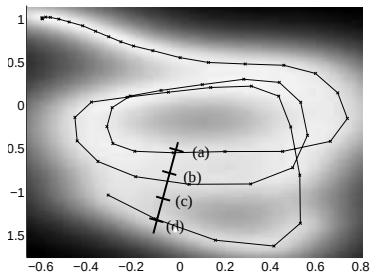


Figure: The latent space for the motion capture data with (*right*) and without (*left*) dynamics. The dynamics use a Gaussian process with an RBF kernel.

Stick Man Results

demStickResults



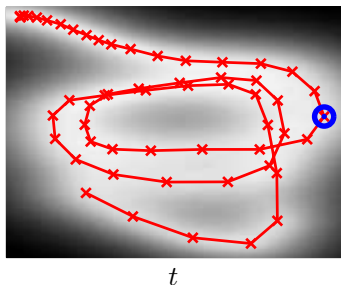
Projection into data space from four points in the latent space. The inclination of the runner changes becoming more upright.

MAP Solutions for Dynamics Models

- Data often has a temporal ordering.
 - ▶ Markov-based dynamics are often used.
 - ▶ For the GP-LVM
 - ★ Marginalising such dynamics is intractable.
 - ★ But: MAP solutions are trivial to implement.
 - ▶ Many choices: Kalman filter, Markov chains *etc.*.
 - ▶ Wang et al. (2006) suggest using a Gaussian Process.

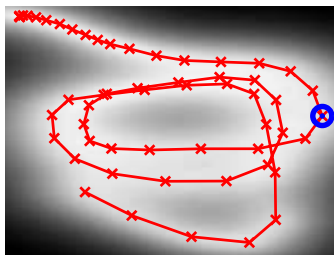
GP-LVM with Dynamics

- Autoregressive Gaussian process mapping in latent space between time points.

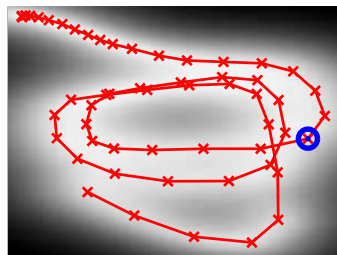


GP-LVM with Dynamics

- Autoregressive Gaussian process mapping in latent space between time points.



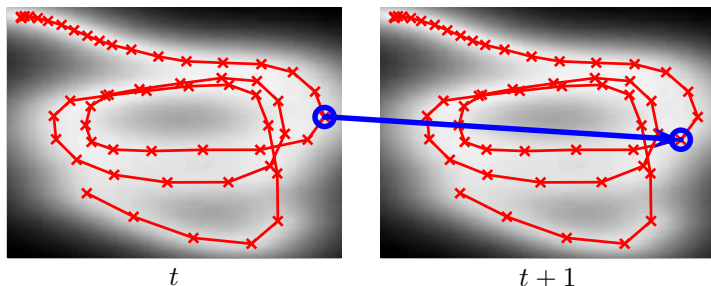
t



$t + 1$

GP-LVM with Dynamics

- Autoregressive Gaussian process mapping in latent space between time points.



demStick1 and demStick2

Figure: The latent space for the motion capture data without dynamics (*left*), with auto-regressive dynamics (*right*) based on an RBF kernel.

Motion Capture Results

demStick1 and demStick2

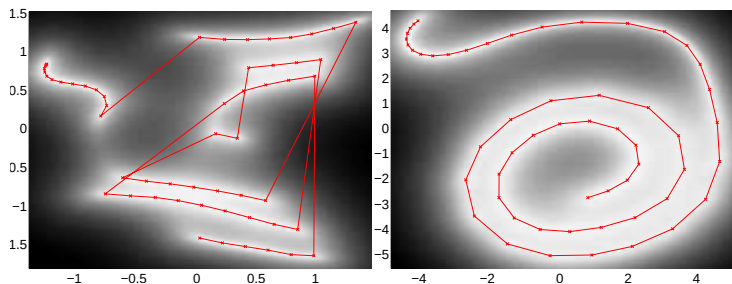
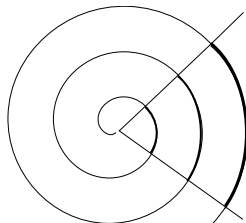


Figure: The latent space for the motion capture data without dynamics (*left*), with auto-regressive dynamics (*right*) based on an RBF kernel.

Inner Groove Distortion

- Autoregressive unimodal dynamics, $p(\mathbf{x}_t | \mathbf{x}_{t-1})$.
- Forces spiral visualisation.
- Poorer model due to inner groove distortion.



Direct use of Time Variable

- Instead of auto-regressive dynamics, consider regressive dynamics.
 - ▶ Take $\mathbf{t}$ as an input, use a prior $p(\mathbf{X}|\mathbf{t})$.
 - ▶ User a Gaussian process prior for $p(\mathbf{X}|\mathbf{t})$.
 - ▶ Also allows us to consider variable sample rate data.

Motion Capture Results

demStick1, demStick2 **and** demStick5

Figure: The latent space for the motion capture data without dynamics (*left*), with auto-regressive dynamics (*middle*) and with regressive dynamics (*right*) based on an RBF kernel.

► Return

Motion Capture Results

demStick1, demStick2 **and** demStick5

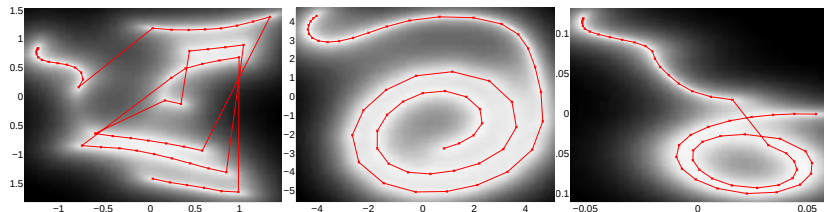


Figure: The latent space for the motion capture data without dynamics (*left*), with auto-regressive dynamics (*middle*) and with regressive dynamics (*right*) based on an RBF kernel.

► Return

Stacking Gaussian Processes

- Regressive dynamics provides a simple hierarchy.
- The input space of the GP is governed by another GP.
- By stacking GPs we can consider more complex hierarchies.
- Ideally we should marginalise latent spaces
 - ▶ In practice we seek MAP solutions.

Two Correlated Subjects

demHighFive1

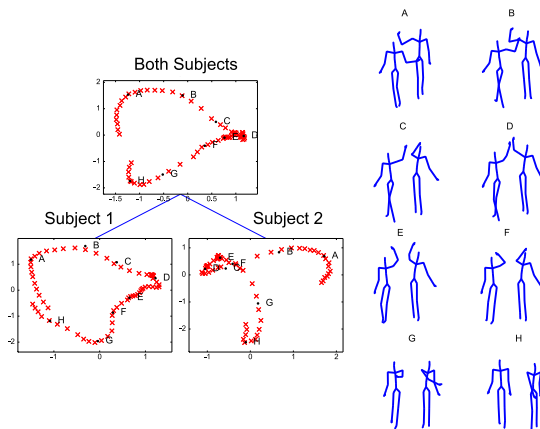


Figure: Hierarchical model of a 'high five'.

Decomposition of Body

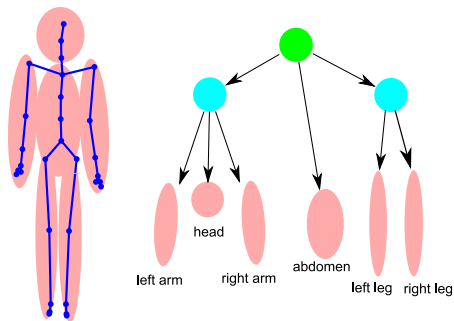


Figure: Decomposition of a subject.

Single Subject Run/Walk

demRunWalk1

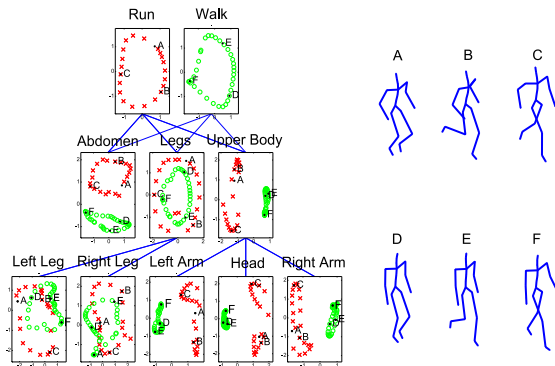


Figure: Hierarchical model of a walk and a run.

► Return

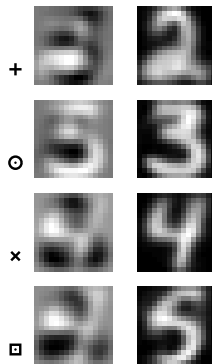
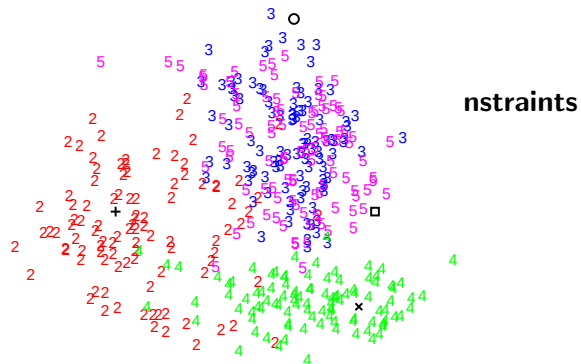
- Special case of back constraints is a *linear* back constraint.

$$\mathbf{X} = \mathbf{YB}$$

where $\mathbf{B} \in \Re^{D \times q}$.

- Maximise the likelihood with respect to the projection matrix $\mathbf{B}$.
- Seems strange to sacrifice the 'non-linearity' of the model in this way.
- Motivate this through a digits data set.

Linear Back Constraints II



Linear Back Constraints III

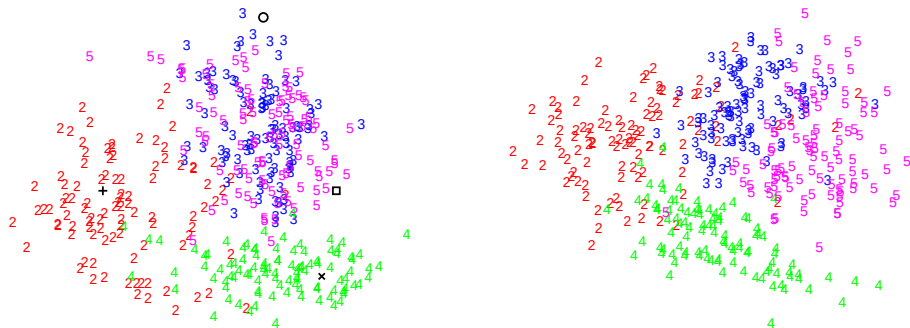


Figure: Linear projections from PCA (*left*) and linear back constrained GP-LVM (*right*)

Comparison for increasing latent dimensionality

q	2	3	4
PCA Errors	131	115	47
Linear constrained GP-LVM Errors	79	60	39

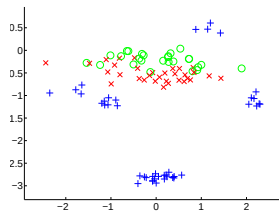
c.f. 24 errors in data space

► Return

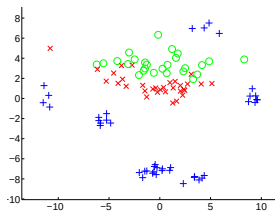
Example Data set

- Oil flow data (Bishop and James, 1993).
 - ▶ Three phases of flow (stratified, annular, homogenous).
 - ▶ Twelve measurement probes.
 - ▶ 1000 data points.
 - ▶ We sub-sampled to 100 data points
 - ▶ Compare, with KPCA, MDS, Sammon mappings, PCA and GTM.

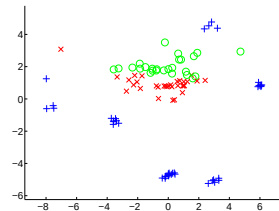
Oil Data II



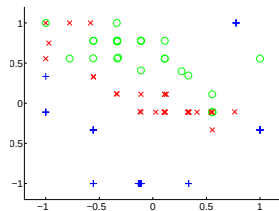
(a) PCA



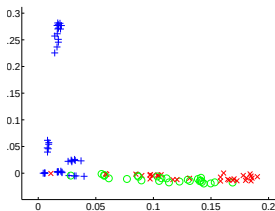
(b) Non-metric MDS



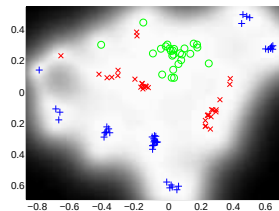
(c) Sammon Mapping



(d) GTM



(e) Kernel PCA



(f) GP-LVM

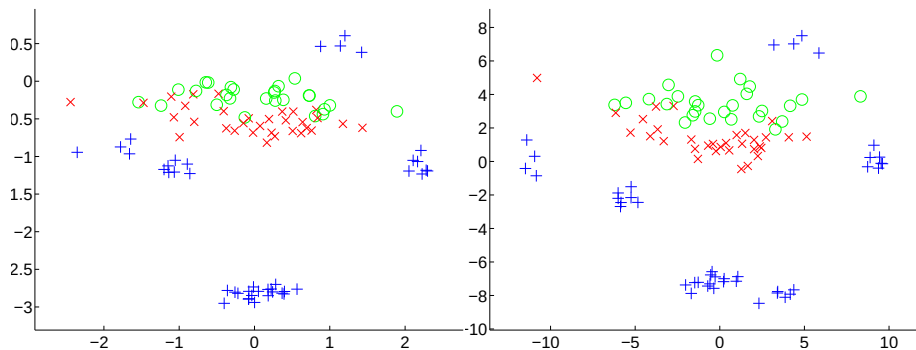


Figure: *Left* PCA, *right* Non-metric MDS

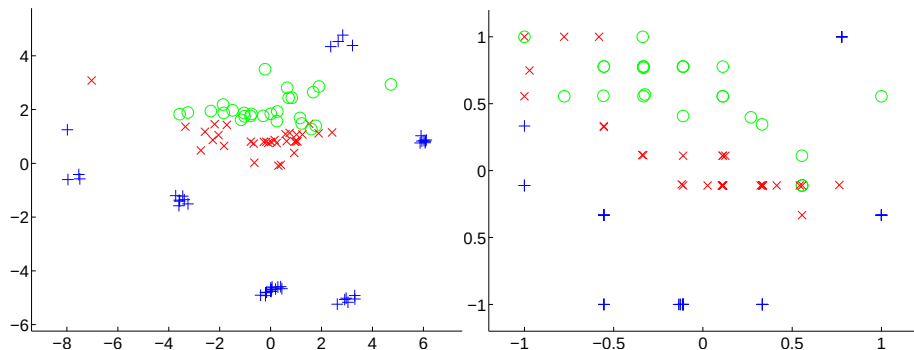


Figure: *Left* Sammon Mapping, *right* GTM

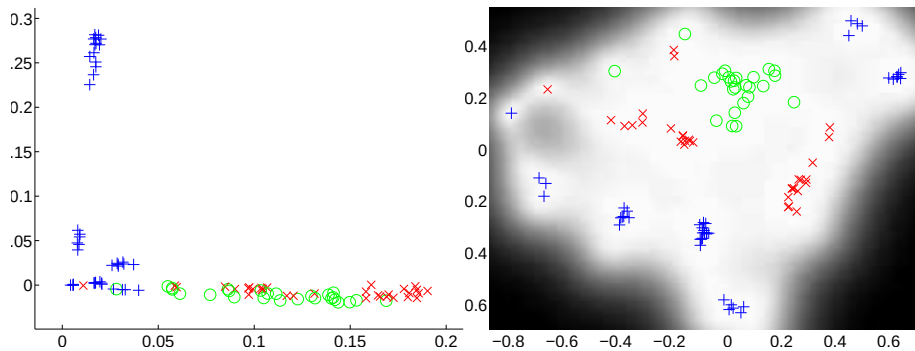


Figure: *Left* Kernel PCA, *right* GP-LVM

Nearest neighbour errors in X space

- Nearest neighbour classification in latent space.

Method	PCA	Non-metric MDS	Sammon Mapping
Errors	20	13	6
Method	GTM*	Kernel PCA*	GP-LVM
Errors	7	13	4

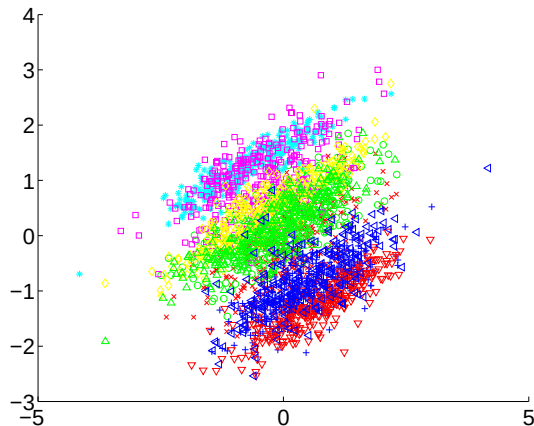
* These models require parameter selection.

► Return

Vocal Joystick Data

- Vowel sounds from a vocal joystick system (Bilmes et al., 2006).
 - ▶ <http://ssli.ee.washington.edu/vj>
- Vowels are from a single speaker and represented as:
 - ▶ cepstral coefficients (12 dimensions) and
 - ▶ 'deltas' (further 12 dimensions).
- 2700 data points in total (300 for each vowel).

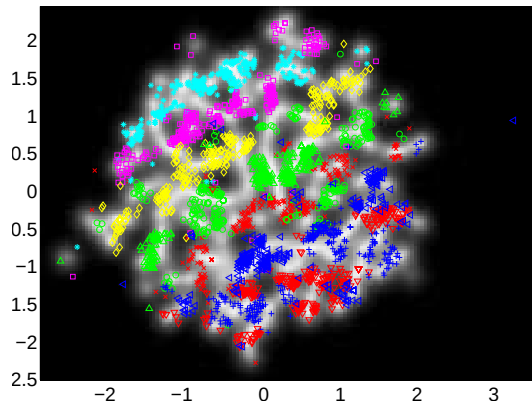
PCA (used as initialisation for GP-LVM)



The different vowels are shown as follows: /a/ red cross /ae/ green circle /ao/ blue plus /e/ cyan asterix /i/ pink square /ibar/ yellow diamond /o/ red down triangle /schwa/ green up triangle and /u/ blue left triangle.

GP-LVM Results

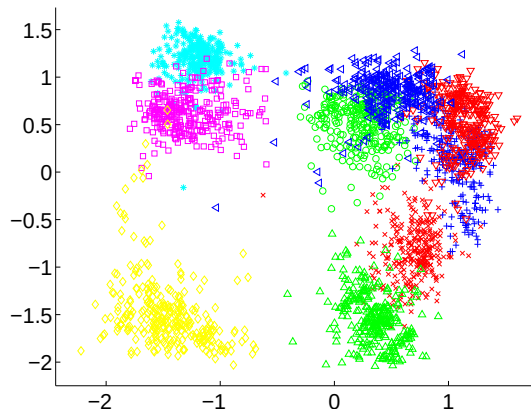
demVowels2



The different vowels are shown as follows: /a/ red cross /æ/ green circle /ao/ blue plus /e/ cyan asterix /i/ pink square /ibar/ yellow diamond /o/ red down triangle /schwa/ green up triangle and /u/ blue left triangle.

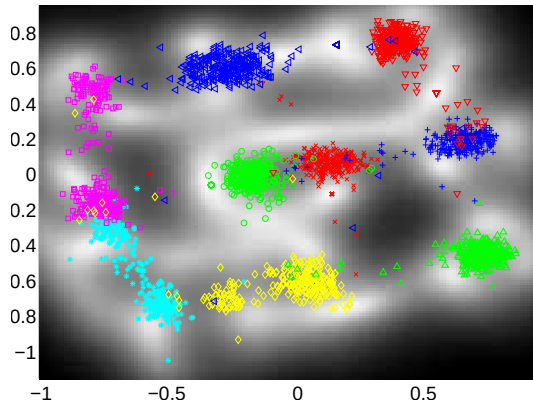
Isomap Results

demVowelsIsomap



The different vowels are shown as follows: /a/ red cross /ae/ green circle /ao/ blue plus /e/ cyan asterix /ibar/ pink square /o/ yellow diamond /schwa/ red down triangle /u/ green up triangle and /u/ blue left triangle.

demVowels3



The different vowels are shown as follows: /a/ red cross /ae/ green circle /ao/ blue plus /e/ cyan asterix /ibar/ pink square /o/ yellow diamond /o/ red down triangle /schwa/ green up triangle and /u/ blue left triangle.

Comparison of the Approaches

- Nearest neighbour classification in latent space.

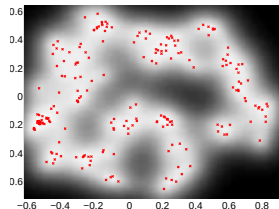
Method	GP-LVM	Isomap	BC-GP-LVM
Errors	226	458	155

cf 24 errors in data space.

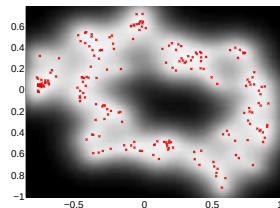
► Return

Navigating by WiFi

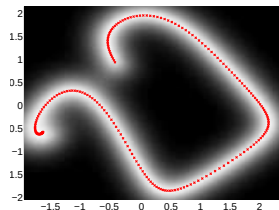
- Wireless access point signal strengths measured by robot moving around building.
- 215 separate signal strength readings.
- 30 separate access points.
- Robot moves in two dimensions so we expect data to be inherently 2-D.
- Learn GP-LVM, GP-LVM with Dynamics, back constrained GP-LVM and back constrained GP-LVM with dynamics.(Ferris et al., 2007)



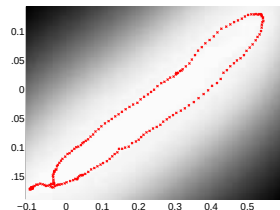
(a) Standard GP-LVM



(b) Standard GP-LVM



(c) Standard GP-LVM



(d) Standard GP-LVM

Figure:

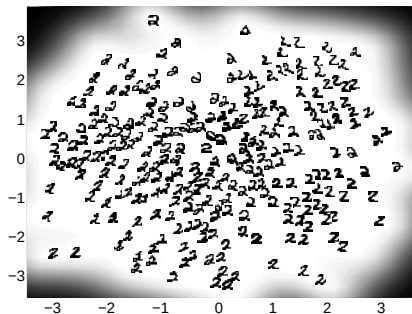
Modelling Binary Data

- A common form of non-Gaussian data is *binary data*.
 - ▶ Can use Assumed Density Filtering to model binary data.
 - ▶ This can also easily be extended to the Expectation Propagation Algorithm Minka (2001).
- Practical consequences:
 - ▶ d times slower.
 - ▶ requires d times more storage.

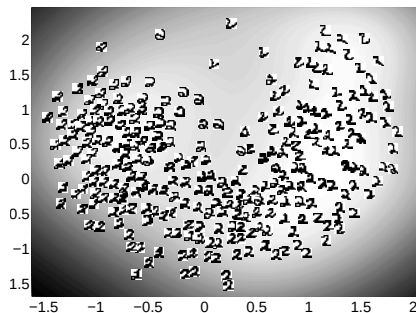
Cedar CD ROM digits

- We model 700 examples of binary 8×8 handwritten twos.
- Use a standard GP-LVM (a Gaussian noise assumption).
- Compare with ADF approximation for the Bernoulli noise model.

Twos Results I



(a) Gaussian Noise Model



(b) Bernoulli Noise Model

Figure:

Two Results II

Reconstruction of Deleted Pixels



Reconstruction Method	Pixel Error Rate
GP-LVM Bernoulli noise	23.5%
GP-LVM Gaussian noise	35.9%
Missing pixels 'not ink'	51.5%

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